ADNOC GAS OPERATIONS & MARKETING LLC

QUALITY REQUIREMENTS FOR EPCM CONSTRUCTION CONTRACTOR

EPCm CONTRACT NO.: 3000076501

ADNOC GAS PROJECT NO.: 30201-50150-LT-HE

LONG TERM - EXECUTE STAGE

PROJECT 5.0 (P5)

EPCm FOR HABSHAN DBN

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Rev								
Revision Date								
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Position	Name	Signature	Name	Signature	Name	Signature	Name	Signature

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1.0 INTRODUCTION

1.1 BACKGROUND

ADNOC aims to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Also, to meet the future oil and gas demand, non-associated gas developments are considered. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. These changes will also subsequently impact the COMPANY downstream facilities, particularly, products handling and transferring facilities i.e., Sales Gas Network, NGL Pipeline Network, Ruwais Fractionation Trains (1, 2, 3, 4 and upcoming new Train 5 part of Bab Cap Gas SELECT Study), Sulphur Handling Facilities, and Condensate systems, etc.

COMPANY has carried out the Long Term DEFINE for the project with M/s Amec International Ltd. (M/s Wood) and subsequently decided to proceed with Long Term EXECUTE phase execution to ensure the readiness of COMPANY facilities to accommodate increased gas supply by December 2027.

Accordingly, COMPANY has awarded EPCm contract to M/s Amec International Ltd. (M/s Wood) to carry out the Long Term scope requirements. M/s Wood shall act as a single entity in the performance of the services and a single point of accountability to ADNOC Gas for the successful completion of Long Term EXECUTE phase.

1.2 LONG TERM EPCM PROJECT OBJECTIVES

The main objective of the Long Term EPCm is to execute the Habshan DBN Project to support ADNOC to achieve oil production target of 5.0 million BOPD so that Associated and Non-Associated Gas can be processed within the overall processing capacity of Habshan. The Long Term EPCm scope includes the following major areas as a minimum:

1. Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in)
2. Bab CDS and Bab NGL
3. Habshan 5
4. Pipeline scope activities including below new pipelines
   I. Bab CDS to Habshan Complex HP pipeline
   II. Bab CDS to Habshan Complex MP pipelines (2 no.s)
   III. Habshan 5 to Habshan 2 pipeline

The scope also includes adequacy checks for Ruwais NGL Fractionation Trains, MERAM,

Habshan Sulphur Granulation Plant, Sales gas, NGL, Condensate and interconnecting Pipeline network and Utilities for all impacted AG facilities.

The Long Term EPCM scope includes the following major activities as a minimum:

1. Detailed Engineering and design services
2. Procurement Services
3. Construction Management services
4. Pre-commissioning, Commissioning, Start-up services
5. Final documentation handover and Project close-out

Project also involves major interface activities with other on-going projects. Refer Project Interface management plan (Doc Num: PP-30201_50150_LT_HE-GG-000-50-00-006 for complete list of interface projects.

1.3 SCOPE BATTERY LIMIT

Refer Design Basis (Doc Num: BOD-30201_50150_LT_HE_GG-000-00-00-001) for the scope battery limits, Gas allocation and detailed Design Basis.

2.0 PURPOSE OF THIS DOCUMENT

The present document is issued by EPCM Contractor to EPCM construction CONTRACTOR as a guideline for implementation of necessary Quality Management System requirements in the performance of the Work for P5 HABSHAN DBN Project and to be used in conjunction with AGES-GL-13-001.

All the requirements set forth in this document shall be considered accepted by the EPCM construction CONTRACTOR and included in its obligations and costs and shall be applied and followed throughout the entire site activities.

EPCM Construction CONTRACTOR Quality System shall fully satisfy all the elements of ISO 9001:2015 “Quality Management System-Requirements” and in addition, an effective mechanism for incorporation of improvement (as per ISO 9004:2018 “Quality Management Systems – guidelines for performance improvement) shall be also included.

EPCM CONSTRUCTION CONTRACTOR Quality Management System implementation shall cover all applicable management aspects for systematic control of all quality related activities performed during construction, pre-commissioning and commissioning up to the final acceptance.

As a general statement, no EPCM CONSTRUCTION CONTRACTOR shall start any work until the relevant QMS and QC procedures are fully issued & approved by EPCM CONTRACTOR & COMPANY with all comments incorporated.

Requirements herein detailed shall be also applicable to Low Tier Subcontractors (Sub-subcontractors).

3.0 ORDER OF DOCUMENT PRECEDENCE

In the event of any conflict between the applicable documents, the order of precedence shall be as follows:

1. UAE Statutory Legislation and Environmental Regulations
2. ADNOC HSE Regulations, HSE Standards and HSE Code of Practice
3. Data Sheets
4. Design Basis
5. Project Specifications and addendum to AGES / DGS (Note-1)
6. COMPANY Standards / Specifications and AGES / DGS (Note-1)
7. International Codes and Standards

Note-1 AGES Addendum/AGES shall be considered wherever AGES addendum/AGES is available. DGS Addendum/DGS shall be considered only in case AGES Addendum/AGES is not available.

However, in case of conflict between the various specifications, standards and the requisition, most stringent requirements shall govern. EPCM CONTRACTOR/VENDOR shall refer any such conflict, and any other matter, which requires further interpretation, to the attention of COMPANY and COMPANY's interpretation shall be final. Resolution of any conflict shall be obtained from the COMPANY in writing before proceeding.

Any deviation from COMPANY standard accepted in the existing facility design shall not be considered as basis for this PROJECT design and any deviations shall be submitted to COMPANY for approval before proceeding.

4.0 REFERENCE, CODES AND STANDARDS

4.1 REFERENCE DOCUMENTS

Table 4-1: Reference Documents

Sl. No.	Document Number	Document Title
1.	PP-30201_50150_LT_HE-GG-000-51-00-001	Project Quality Plan
2.	PP-30201_50150_LT_HE-GG-000-51-00-003	Non-conformance Procedure

Sl. No.	Document Number	Document Title
3.	PP-30201_50150_LT_HE-GG-000-51-00-004	Construction Quality Plan
4.	PP-30201_50150_LT_HE-GG-000-51-00-008	SITE INSPECTION AND COORDINATION PROCEDURE
5.	30201_50150_LT_HE-ADD-SP- 01-002	Addendum to AGES-SP-01-002 Structural Steel Supply, Fabrication and Erection
6.	30201_50150_LT_HE-ADD-SP- 09-007	Addendum to AGES-SP-09-007 Specification for Shop and Field fabrication of Piping
7.	30201_50150_LT_HE-ADD-SP- 09-005	Addendum to AGES-SP-09-005 Specification for Gasket and Fasteners
8.	30201_50150_LT_HE-ADD-SP- 09-006	Addendum to AGES-SP-09-006 Pipe supports Specification
9.	30201_50150_LT_HE-ADD- 1300-016	Addendum to DGS-1300-016 piping color identification specification
10.	30201_50150_LT_HE-ADD-SP- 07-004	Addendum to AGES-SP-07-004 painting and coating Specification
11.	30201_50150_LT_HE-ADD- SP-07-007	Addendum to AGES-SP-07-007 Welding & Non- Destructive Examination (NDE)
12.	30201_50150_LT_HE-ADD- SP-07-003	Addendum to AGES-SP-07-003 Requirements for Materials in Severe Service
13.	30201_50150_LT_HE-ADD- SP-09-014	Addendum to AGES-SP-09-014 Fibre Reinforced Plastic Piping and Pipeline System
14.	30201_50150_LT_HE-ADD- SP-07-002	Addendum to AGES-SP-07-002 External Pipeline Coatings Specification
15.	30201_50150_LT_HE-ADD-6300-004	Addendum to DGS-6300-004 pipeline welding specification
16.	30201_50150_LT_HE-ADD-SP-10-014	ADDENDUM TO AGES-SP-10-014 PIPELINE CLEANING, GAUGING, FILLING AND

Sl. No.	Document Number	Document Title
		HYDROSTATIC TESTING
17.	30201_50150_LT_HE-ADD-SP-10-003	Addendum to AGES-SP-10-003 Onshore Pipelines Design and Construction Specification
18.	30201_50150_LT_HE-ADD-SP-09-005	ADDENDUM TO AGES-SP-09-005 SPECIFICATION FOR GASKETS AND FASTENERS
19.	30201_50150_LT_HE-ADD-SP-09-008	ADDENDUM TO AGES-SP-09-008 INSULATION SPECIFICATION
20.	QUA-PCM-100011	Control of Non-Conformance Process Map
21.	PP-30201_50150_LT_HE-GG-000-51-00-002	Audit procedure
22.	30001-50150G-ADD-SP-01-001	ADDENDUM TO AGES-SP-01-001 CONCRETE SUPPLY AND CONSTRUCTION
23.	30201_50150_LT_HE-ADD-GL-01-001	Addendum to AGES-GL-01-001- Assessment and Structural Adequacy Check of Existing Structures and Pipe Racks
24.	30201_50150_LT_HE-ADD-SP-01-004	Addendum to AGES-SP-01-004- Anchor Bolts and Grouting Specification
25.	30201_50150_LT_HE-ADD-SP-01-005	Addendum to AGES-SP-01-005- Roads and Paving - Design and Construction Specification
26.	30201_50150_LT_HE-ADD-SP-01-006	Addendum to AGES-SP-01-006- Site Preparation, Earthworks and Bund Construction Specification
27.	30201_50150_LT_HE-ADD-SP-01-009	Addendum to AGES-SP-01-009- Civil Construction Materials Sampling and Testing Specification
28.	30201_50150_LT_HE-ADD-SP-01-012	Addendum to AGES-SP-01-012- Concrete Repair
29.	30201_50150_LT_HE-ADD-SP-07-001	ADDENDUM TO AGES-SP-07-001 CATHODIC PROTECTION SPECIFICATION

Sl. No.	Document Number	Document Title
30.	30201_50150_LT_HE-ADD-SP-02-011	ADDENDUM TO AGES-SP-02-011 ELECTRICAL POWER, CONTROL AND EARTHING CABLES SPECIFICATION
31.	30201_50150_LT_HE-ADD-SP-02-019	ADDENDUM TO AGES-SP-02-019 ELECTRIC BULK MATERIALS SPECIFICATION
32.	30201_50150_LT_HE-ADD-SP-02-020	ADDENDUM TO AGES-SP-02-020 LIGHTING AND SMALL POWER DISTRIBUTION BOARDS SPECIFICATION
33.	30201_50150_LT_HE-ADD-SP-14-003	ADDENDUM TO AGES-SP-14-003 HVAC DUCTWORK, PIPEWORK AND ACCESSORIES
34.	30201_50150_LT_HE-ADD-SP-14-002	ADDENDUM TO AGES-SP-14-002 HVAC EQUIPMENT SPECIFICATION
35.	30201_50150_LT_HE-ADD-SP-07-006	ADDENDUM TO AGES-SP-07-006 FIELD JOINT COATING FOR PIPELINE
36.	30201_50150_LT_HE-ADD-SP-14-002	ADDENDUM TO AGES-SP-14-002 HVAC EQUIPMENT SPECIFICATION
37.	30201_50150_LT_HE-ADD-SP-14-003	ADDENDUM TO AGES-SP-14-003 HVAC DUCTWORK, PIPEWORK AND ACCESSORIES
38.	PMG-PRO-110131	LESSONS LEARNED PROCEDURE
39.	SP-30201_50150_LT_HE-GG-000-13-00-001	SPECIFICATION FOR BOLT TORQUING TENSIONING PROCEDURE FOR FLANGED CONNECTIONS
40.	30201_50150_LT_H7-H7-700-MP-ADD-00003	ADDENDUM TO AGES-SP-09-003 MANUAL PIPING AND PIPELINE VALVES
41.	PP-30201_50150_LT_HE-GG-000-69-00-005	GOLDEN WELD PROCEDURE
42.	SOW-30201_50150_LT_HE-GG-000-51-00-002	PIPING SOW for FIELD TESTING OF VALVES

4.2 ADNOC STANDARDS

Table 4-2: ADNOC Standards

Sl. No.	Document Number	Document Title
1.	AGES-GL-13-001	Contractors QA/QC Requirement
2.	AGES-SP-13-002	Procurement Inspection and Certification Requirement in Projects
3.	AGES-GL-13-002	Positive material specification
4.	AGES-SP-13-003	Traceability of Shop & Field Piping Materials
5.	Ages-SP-13-001	Criticality rating specification
6.	AGES-SP-01-001	Concrete Supply and Construction Specification
7.	AGES-SP-01-002	Structural Steel Supply, Fabrication and Erection Specification
8.	AGES-SP-01-003	Structural Design Basis - Onshore Specification
9.	AGES-SP-01-004	Anchor Bolts and Grouting Specification
10.	AGES-SP-01-005	Roads and Paving - Design and Construction Specification
11.	AGES-SP-01-006	Site Preparation, Earthworks and Bund Construction Specification
12.	AGES-SP-01-007	Drainage Systems - Design and Construction Specification
13.	AGES-SP-01-009	Civil Construction Materials Sampling and Testing Specification
14.	AGES-SP-01-012	Concrete Repair
15.	AGES-SP-01-013	Architectural Design Basis
16.	AGES-SP-01-014	Architectural Finishes and Furnishing

Sl. No.	Document Number	Document Title
17.	AGES-SP-01-015	Architectural Specialties, Metal, Wood and Plastics
18.	AGES-SP-01-016	Doors and Windows
19.	AGES-SP-01-019	Prefabricated Metal Buildings
20.	AGES-SP-01-021	Miscellaneous Onshore Civil Works and Demolition
21.	AGES-SP-01-022	Fence and Gate
22.	AGES-SP-01-029	Landscaping
23.	AGES-GL-01-001	Assessment and Structural Adequacy Check of Existing Structures and Pipe Racks
24.	AGES-SP-02-019	Electrical Bulk Material Specification
25.	AGES-SP-02-020	Lighting and Small Power Distribution Boards
26.	AGES-PH-02-002	Lighting and Small Power
27.	AGES-SP-04-011	Fire & Gas Detectors and Related Devices
28.	AGES-SP-07-001	Cathodic Protection Specification
29.	AGES-SP-07-002	External Pipeline Coatings Specification
30.	AGES-SP-07-003	Requirement for Materials in Severe Service
31.	AGES-SP-07-004	Painting & Coating Specification
32.	AGES-SP-07-005	Requirements for Cr-Mo / Cr-Mo-V Steel in High Temperature, High Pressure Hydrogen Service
33.	AGES-SP-07-006	Field Joint Coating for Line Pipe
34.	AGES-SP-07-007	Welding & Non-Destructive Examination (NDE)
35.	AGES-SP-07-009	Galvanizing
36.	AGES-SP-07-010	Internal Cleaning of Piping System Specification

Sl. No.	Document Number	Document Title
37.	AGES-SP-07-012	Chemical Treatment and Corrosion Monitoring System Specification
38.	AGES-SP-07-011	Preservation & Export Packing Specification
39.	AGES-SP-09-005	Specification for Gaskets and Fasteners
40.	AGES-SP-09-006	Pipe Support Specification
41.	AGES-SP-09-009	Piping Tie-ins & Hot Tapping
42.	AGES-SP-09-014	Fibre Reinforced Plastic Piping and Pipeline Systems
43.	AGES-SP-10-003	Onshore Pipelines Design and Construction Specification
44.	AGES-SP-10-014	Pipeline Cleaning, Gauging, Filling and Hydrostatic Testing Specification
45.	AGES-SP-14-002	HVAC Equipment Specification
46.	AGES-SP-14-003	HVAC Ductwork, Pipework and Accessories
47.	AGES-SP-14-005	Testing, Adjusting and Balancing of HVAC Specification
48.	AGES-SP-14-004	HVAC Control Specification

4.3 INTERNATIONAL CODES AND STANDARDS

Table 4-3: International Codes and Standards

Sl. No.	Document Number	Document Title
1.	ISO 9000:2015	Quality management systems – Fundamentals and vocabulary
2.	ISO 9001:2015	Quality Management System Requirements
3.	ISO 9004:2018	Guidance to achieve sustained success

Sl. No.	Document Number	Document Title
4.	ISO 10005:2018	Guideline for Quality Plan
5.	ISO 10006:2017	Guidelines for Quality Management in Projects
6.	ISO 19011:2018	Guidelines for Auditing Management Systems

5.0 DEFINITIONS

5.1 GENERAL DEFINITIONS

For this document, the following general definitions are applicable.

Table 5-1: General Definitions

DEFINITION	DESCRIPTION
COMPANY	ADNOC GAS (AG) for whom the SERVICES are being performed by ENGINEERING CONTRACTOR.
EPCM CONTRACTOR	Amec International Limited – Abu Dhabi (WOOD) appointed by COMPANY to perform Engineering, Procurement, Construction management, Pre-Commissioning and Commissioning stage for Long Term Solution (under EXECUTE phase).
FACILITIES	Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in), Bab CDS, Bab NGL, Habshan 5 and interconnecting pipeline network.
LICENSOR	Supplier of packaged units and responsible for the operating and design performance.
MANUFACTURER/ VENDOR/ SUPPLIER	The Party (parties) which manufactures and/or supplies material, equipment, technical documents or drawings and/or services to perform the duties specified by the COMPANY
PMC	Project Management Consultant appointed by COMPANY for the PROJECT
PMT	Project Management Team
PROJECT	PROJECT 5.0 (P5), Long Term EPCm for HABSHAN DBN
SERVICES	The scope of work to be completed during the EXECUTE

DEFINITION	DESCRIPTION
	stage (EPCm) of the PROJECT.
SHALL	The use of the word “shall” indicate a mandatory requirement
SHOULD	The use of the word “should” indicate a strong recommendation to comply with the requirements of this document
EPCM CONSTRUCTION CONTRACTOR	Subcontractor appointed by EPCm CONTRACTOR to carry out Construction works at site for respective construction packages.

5.2 TECHNICAL DEFINITIONS

For this document, the following technical definitions are applicable.

Table 5-2:Technical Definitions

DEFINITION	DESCRIPTION
Acid Gas	A gas stream containing predominantly H_2S and/or CO_2 from an Acid Gas Removal Unit
Condensate	A hydrocarbon liquid stream containing Pentane+ compounds
Dehydrated Gas	A gas stream that has been dehydrated to remove water
Dry Gas	A gas stream containing no liquids at flowing conditions, but which may still contain water and hydrocarbons that will fall out if there is a change in flowing conditions.
Feed Gas - Associated Gas	A Gas stream that is produced from an oil well during the extraction of crude oil.
Feed Gas – Non-associated Gas	Hydrocarbon gas originating from gas reservoirs.
GMP 2023	Gas Master Plan 2023
Lean Gas	Lean nature of gas is defined based C2+ in the gas stream. If C2+ is less than 15% then stream is defined as Lean gas
Mixed Feed Gas Stream	Mixed associated and non-associated gas stream

DEFINITION	DESCRIPTION
NGL	A liquid stream with primarily C2, C3, C4 and some C5+ compounds.
Rich Gas	Rich gas is defined based C2+ in the gas stream. If C2+ is more than 15% it is considered rich gas
Sales Gas	Residue gas sent to the Sales gas network.
Sour Gas	A gas stream containing significant amounts of H_2S and/or CO_2.
Sustainable Profile (GMP 2023)	GMP 2023 High Supply Scenario – unconstrained oil production from GMP Guidelines Document from all associated gas fields
Sweet Gas	A gas stream in which the H_2S and CO_2 are not present in significant amounts.
Technical Profile	AON provided profile for technical case for associated gas.
Technical Optimized case	The ADNOC Gas facilities (Das, Asab-0/3, Buhasa, Ruwais, associated pipelines) will be sized to accommodate technical rates from associated gas fields. Habshan facilities will be sized to accommodate technical rate from all associated gas fields (Bab, Buhasa, NEB, Offshore) but not simultaneously. All inlet facilities / inlet compression shall be designed for technical rate.
Unconstrained Production Profile / High Supply Scenario	This shall be based on Upstream's OPEC Unconstrained Oil production case. This scenario corresponds to sustainable projects profiles

5.3 ABBREVIATIONS

ADNOC	Abu Dhabi National Oil Company (COMPANY)
AG	ADNOC Gas
AGES	ADNOC General Engineering Specifications

BMS	Business Management System
BOPD	Barrels Oil Per Day
CAPA	Corrective and Preventive Actions
CAR	Corrective Action Register
EPCM CC	EPCM CONSTRUCTION CONTRACTOR
CRS	Comment resolution Sheet
DCA	Design Concession Authorization
DGS	Design General Specification
EDDR	Engineering Documents Deliverable Register
EDMS	Engineering Data Management System
ENVID	Environment HAZID
EPC	Engineering Procurement & Construction
FEED	Front End Engineering & Design
GEC	Global Execution Centre
H2S	Hydrogen Sulphide
HAZID/ENVID/OHID	Hazard Identification
HAZOP	HAZID & Operability
HSE	Health Safety Environment
HSEQ	Health, Safety and Environmental and Quality
IDC	Inter Disciplinary Check
IFA	Issued For Approval
IFD	Issued For Design
IFE	Issued For Enquiry

IFH	Issued For HAZOP
IFI	Issued For Information
IFP	Issued For Purchase
IFR	Issued For Review
IFT	Issued For Tender
IGD	Integrated Gas Development Project
ISO	International Organization for Standardization
KPI	Key Performance Indicator
LLI	Long Lead Items
LOA	Letter of Award / Letter of Assistance
MOC	Management of Change
MOM	Minutes of Meeting
MMBOPD	Millions of Barrels of Oil Per Day
NCR	Non-Conformance Report
NGL	Natural Gas Liquids
AG	Offshore Associated Gas
OEM	Original Equipment Manufacturer
OHID	Occupational Health HAZID
PDCA	Plan Do, Check, Act
PMT	Project Management Team
PQP	Project Quality Plan
QA/QC	Quality Assurance and Quality Control
SOW	Scope of Work

TBE	Technical Bid Evaluation
TQ	Technical Query
VAP	Value Assurance Process
WBS	Work Breakdown Structure
UAE	United Arab Emirates
VAP	Value Assurance Process

6.0 EPCM CONSTRUCTION CONTRACTOR RESOURCES

The EPCM CONSTRUCTION CONTRACTOR shall have an independent Quality Control organization with sufficient resources to oversee the SERVICES. All Quality Control personnel shall be independent of any related production or construction function EPCM CONSTRUCTION CONTRACTOR quality personnel assigned to the PROJECT must be technically competent to perform duties assigned. Records documenting qualification and references of EPCM CONSTRUCTION CONTRACTOR quality personnel must be maintained for EPCM CONTRACTOR and COMPANY review. All variations not agreed and formalised with EPCM CONTRACTOR and COMPANY are on EPCM CONSTRUCTION CONTRACTOR responsibility & charge.

All QA and QC personnel employed by EPCM CONSTRUCTION CONTRACTOR shall have reporting lines through EPCM CONSTRUCTION CONTRACTOR Quality Management organization.

EPCM CONSTRUCTION CONTRACTOR shall submit to EPCM CONTRACTOR an Organization Chart with all planned Quality positions and personnel nominees.

The EPCM CONSTRUCTION CONTRACTOR shall submit the minimum number of quality resources that are required for the execution of the project. EPCM CONSTRUCTION CONTRACTOR shall submit for EPCM CONTRACTOR & COMPANY review a detail quality resource planning and their mobilization plan for construction phase including shop, fabrication yards and site inspection & pre-commissioning.

Before the start of the work, the EPCM CONSTRUCTION CONTRACTOR shall submit to the EPCM CONTRACTOR/COMPANY for approval the names of the personnel appointed by EPCM CONSTRUCTION CONTRACTOR to perform the inspection and testing activities (the QC Personnel) together with their curriculum vitae and qualification certificates. EPCM CONSTRUCTION CONTRACTOR QC Personnel shall be independent from the one that has supervised or directly performed the work (independent from their construction team).

EPCM CONTRACTOR has the authority to interview the EPCM CONSTRUCTION

CONTRACTOR proposed QA/QC, all discipline Inspectors, NDT & any EPCM CONSTRUCTION CONTRACTOR candidates involved in execution assurance activities and could also disqualify any individual on unsatisfactory performance. COMPANY also has the right to request interview of any Key QA/QC focal point from EPCM CONSTRUCTION CONTRACTOR.

EPCM CONTRACTOR shall have at any time, the right to request the replacement of any member of the EPCM CONSTRUCTION CONTRACTOR QC Personnel who, at the EPCM CONTRACTOR and/or COMPANY sole discretion, when the candidate demonstrated incompetence or negligence, and the EPCM CONSTRUCTION CONTRACTOR shall promptly replace the member at his own cost. EPCM CONSTRUCTION CONTRACTOR to ensure that replaced resource will have as minimum same or higher level of competency, experience/knowledges required to cover the position in question.

EPCM CONSTRUCTION CONTRACTOR site QA/QC manager shall be mobilized for all duration of the site work through final acceptance activities before one month before start of EPCM CONSTRUCTION CONTRACTOR activities.

In case of vacations, illness, temporary assignments, emergency leave, resignations, or other reasons, as necessary, EPCM CONSTRUCTION CONTRACTOR shall officially notify EPCM CONTRACTOR at least once moth before and shall maintain the competent replacements for the QA/QC personnel to maintain required QA/QC coverage. In Appendix 1 are detailed the Minimum Requirements for Number of Quality Personnel, as per PROJECT requirements.

EPCM CONSTRUCTION CONTRACTOR personnel appointed for Quality Control, NDT, PMI and welding; painting & corrosion activities shall be qualified as per PROJECT requirements detailed in Appendix 2 & AGES-GL-13-001. EPCM CONTRACTOR and COMPANY shall review and approve such qualifications prior to the start of specific activities.

QC Personnel qualification records shall be kept updated by the EPCM CONSTRUCTION CONTRACTOR for the entire duration of the works.

EPCM CONSTRUCTION CONTRACTOR must seek EPCM CONTRACTOR & COMPANY review and approval to subcontract services to Independent Third-Party Agencies. These services include inspection, testing, certification, calibration, NDT, audits, and technical consultancy.

EPCM CONSTRUCTION CONTRACTOR shall submit a listing of QA/QC personnel, sorted by discipline for each WORK Phase, to EPCM CONTRACTOR & COMPANY Representative within thirty (30) calendar days of the effective date of the CONTRACT and shall update and submit the list monthly (with changes noted).

EPCM CONSTRUCTION CONTRACTOR shall mobilize the approved QA/QC personnel ten (10) working days prior to the start-up of the construction activities of their discipline. This period is intended for the inspector to familiarize with the AGREEMENT scope of SERVICES

as well as the Standards and Specifications, in addition to having a formal interview with EPCM CONTRACTOR & COMPANY Representative. EPCM CONSTRUCTION CONTRACTOR shall update this Organization Chart & deployment plan minimum monthly and issue it to EPCM CONTRACTOR & COMPANY.

7.0 EPCM CONSTRUCTION CONTRACTOR QUALITY RESPONSABILITIES

EPCM CONSTRUCTION CONTRACTOR shall implement a Quality Management System according to the ISO 9001:2015 (Quality Management Systems – Requirements) and ISO 9004:2018 (Quality Management Systems – Guidelines for Performance Improvement).

EPCM CONSTRUCTION CONTRACTOR shall submit the Project specific Quality plan where, EPCM CONSTRUCTION CONTRACTOR's Project Quality Plan according to ISO 10005:2018 & AGES-GL-13-001 requirements to be submit for EPCM & COMPANY review and approval.

Prior to the start of construction activity, EPCM CONTRACTOR shall initiate a quality preproduction meeting with EPCM CONSTRUCTION CONTRACTORS to ensure full understanding on the execution SOW, quality requirements and the level of required control on the construction activities.

EPCM CONSTRUCTION CONTRACTOR shall be directly responsible of the control of its construction SOW activities.

EPCM CONTRACTOR will manage & monitor EPCM CONSTRUCTION CONTRACTOR work execution, performing inspections in accordance with the approved ITPs and witness all work and tests performed by EPCM CONSTRUCTION CONTRACTOR.

EPCM CONSTRUCTION CONTRACTOR shall provide software containing Weld history, NDT history (hyperlinking to all NDE reports and also to the uncompressed RT image), Hardness, PWHT History, MTC for all sites and shop piping weld joints. This Software shall also contain Unit System, Sub system and Isometric details for each of the welds. Further, MTC no's and Heat no. for the adjoining piping components shall also be provided.

EPCM CONSTRUCTION CONTRACTOR can choose the Software with the proven track record for ADNOC Projects. Further, EPCM CONSTRUCTION CONTRACTOR shall present/ Demonstrate Software to COMPANY Personnel for review and approval prior to implementation.

EPCM CONSTRUCTION CONTRACTOR shall submit to EPCM CONTRACTOR on weekly basis an inspection three- weeks look ahead schedule of all planned QA/QC activities for EPCM CONSTRUCTION CONTRACTOR's procurement (as applicable), construction, and pre-commissioning work phase.

8.0 EPCM CONSTRUCTION CONTRACTOR QUALITY REQUIREMENT

8.1 General

EPCM CONSTRUCTION CONTRACTOR shall provide for review and approval the below listed documentation, well before construction activities, within the defined timing, and as applicable, according to specific scope of work.

All procedures and personnel qualification shall be submitted for review and/or approval at least thirty (30) calendar days prior to the start of the work.

The following table summarizes deliverables under EPCM CONSTRUCTION CONTRACTOR responsibility to be submitted to EPCM CONTRACTOR & COMPANY for approval and for periodical reporting:

Title	Submittal to CONTRACTOR for approval	Periodical reporting
EPCM CONSTRUCTION CONTRACTOR Quality Plan	15 days from award	
Nonconformity management procedure (as per ISO 9001 requirements)	One month from award	
Quality Records management procedure	One month from award	
Project document management procedure	Within 15 days from the award.	
Audit Management procedure (as per ISO 9001 & 19011 requirements)	One month from award	
Procedure for Calibration of Measuring Equipment	One month from award	
Procedure for Filing documentation	One month from award	
Procedure for material handling	One month from award	
Site Preservation Procedures	One month from award	
QC Procedures discipline wise	One month from award	
QA/QC Personnel Qualification	15 days from award	
Quality audit schedule	One month from award	

EPCM CONSTRUCTIONCONTRACTOR inspection and test plans (discipline/activity wise)	One month from award	
Site inspection coordination procedure	One month from award	
Inspection three-weeks look ahead schedule of all planned QA/QC activities		On weekly basis
Non-conformity report	Within three (3) calendar days	
NCR summary		On weekly basis
Inspection / fabrication status report		On weekly basis
RFI logbook		On weekly basis
Procedures but not limited according to scope for refractory installation, welding, welding repair, NDT, heat treatment, PWHT and alloy verification (PMI), painting, coating, etc..	One month from award	
Early work and civil execution procedures.	One month from award	
Mechanical erection and installation procedures	One month from award	
Fabrication procedures.	One month from award	
Procedure for Welding consumable	One month from award	
Welding Procedure Qualification Records (PQR)	One month from award	
Qualification of PWPS	One month from award	
Welding Procedure Specifications (WPS)	One month from award	
Welding & Log		On weekly basis
Qualifications of welders against WPSs	One month from award	

Welder Performance Qualification Record and relevant Welder List	One month from award	
Non-Destructive Test (NDT) Personnel List with the relevant certification and Curriculum Vitae	One month from award	
Weld production (welder performance, productivity of each welder and welding groups, weld reject rates, and NDT status including status of weld repairs & NDT back log		On weekly basis
Method statements covering performing activities	One month from award	
Flange management procedure	Before starting activity with one month at least	
Preservation cards/check sheets		According to frequency prescribed for preservation
Lower their Subcontractor's ITPs	One month from award	
Final Dossier/Documentation Issuing Program (Final Data Book)	Preliminary Programmed at the KOM. Final Program reached 50% of work progress	Updating On monthly bases

This is to be considered a preliminary list of site procedures will be finalized during the construction phase with the appointed EPCM CONSTRUCTION CONTRACTOR against the SOW, additional procedures could be added according to the project need and SOW.

EPCM CONSTRUCTION CONTRACTOR is not allowed to start any work until the relevant QMS and QC procedures are fully approved, or all comments are incorporated.

Responsibility matrix for main tentative documents issuance added in Appendix 4.

8.2 SITE/CONSTRUCTION PROJECT QUALITY PLAN

EPCM CONSTRUCTION CONTRACTOR shall prepare and submit its Site/Construction project Quality Plan within 15 days from the effective date of the Subcontract.

Site Quality Plan shall include:

Narrative details of how the EPCM CONSTRUCTION CONTRACTOR interprets and implements ISO 9001:2015 requirements for the PROJECT SCOPE OF WORK

A matrix format detailing how the EPCM CONSTRUCTION CONTRACTOR's quality system addresses all the elements of ISO 9001:2015 & AGES-GL-13-001.

Moreover, Site/Construction Quality Plan shall contain an EPCM CONSTRUCTION CONTRACTOR organization chart, detailing its organization responsible for performing the work and showing the assignments and numbers of personnel responsible for the quality control activities.

The Site/construction Quality Plan shall comply with the contract documents and with the COMPANY quality requirements, project specifications and International Standard specified referenced within the contract. Particular attention shall be given but not limited to the following main topics as categorized by the international standard ISO 9001:2015 & AGES-GL-13-001:

- Risk-based thinking
- Management Responsibility
- Resource Management
- Service and product realization
- Measurement, analysis, and improvement

If the EPCM CONSTRUCTION CONTRACTOR scope of work entails engineering, purchasing, and construction, then the Quality Plan shall address all three areas. If the EPCM CONSTRUCTION CONTRACTOR work is limited to construction mechanical completion and Pre-commissioning, then the Quality Plan shall address only those areas.

EPCM CONSTRUCTION CONTRACTOR shall develop Key Performance Indicators (KPIs) Covering his Construction SOW and in process divisions (Project management, Quality management, progress/schedule, construction) to measure how well the executions integrity & objectives are being met. Where the KPI data shows that the objectives are not being systematically achieved, EPCM CONSTRUCTION CONTRACTOR Project Management Team will investigate the causes and agree and implement corrective action with immediate actions and shall be agreed with EPCM CONTRACTOR & COMPANY those KPIs shall be as per the AGES-GL-13-001 in in line with the Project KPIs.

8.3 SITE INSPECTION & TEST PLAN

One month prior to the start of the applicable WORK, EPCM CONSTRUCTION CONTRACTOR shall submit for review and approval a detailed Inspection and Test Plan (ITP) for all construction processes to be executed by EPCM CONSTRUCTION CONTRACTOR or their SUBCONTRACTORs (including Precast & fabrication yards).

Site Inspection & test Plans are prepared by EPCM CONSTRUCTION CONTRACTOR (according to specific discipline) and will be submitted to EPCM CONTRACTOR /COMPANY for review and approval.

EPCM CONSTRUCTION CONTRACTOR ITPs shall contain information of its own documents developed for the PROJECT (e.g. Method statement, procedures, calibration procedure, other required documentation to carry on their activities).

If the contract requires the EPCM CONSTRUCTION CONTRACTOR to provide for inspection of purchased materials and equipment, the EPCM CONSTRUCTION CONTRACTOR shall provide the same competent inspection as would be required by EPCM CONTRACTOR.

EPCM CONSTRUCTION CONTRACTOR performing construction and Pre-commissioning work shall implement approved inspection and test plan pertinent to the work they will be performing, stating & against all project applicable reference and acceptable procedures.

EPCM CONTRACTOR and the COMPANY shall have the right to inspect or test any part of the WORKS, as they deem appropriate. Therefore, all inspections and tests for which the presence of the EPCM CONTRACTOR and the COMPANY required are marked in the relevant column of the Inspection & Test Plans with the code “W: Witness” or “H: Hold”, or “WR: Witness Random”, “M” Monitor.

As a minimum each ITP shall include the following information:

1) Process Activity description
2) Quality control requirements
3) Responsibilities
4) Applicable procedures / Codes and standards
5) Acceptance criteria / Extend of Inspection
6) Percentage of tests to be done (including increased inspection levels when performance requirements are not met)
7) Verifying documents/ Records
8) Review Witness and Hold points.

All Quality Control (QC) activities and listing status/records shall be compiled on a database referencing the appropriate ITP, procedure, and certification so that the QC status can be continually monitored as work progresses.

EPCM CONSTRUCTION CONTRACTOR shall invite to inspection EPCM CONTRACTOR and COMPANY according to what they have indicated as intervention points (“H” – “W” – “WR”) via formal request for inspection “RFI” issued discipline wise covering inspection activities daily refer to PP-30201_50150_LT_HE-GG-000-51-00-008 site inspection & coordination procedure.

EPCM CONSTRUCTION CONTRACTOR shall implement a robust Inspection notification and inspection records, and shall be digitalized, all inspection records need to be traceable and recorded on electronic/digital document management system.

EPCM CONSTRUCTION CONTRACTOR shall issue a site inspection coordination procedure for EPCM CONTRACTOR/COMPANY review and approval, which shall follow Project specification and AGES-GL-13-001 requirements addressing the handling, coordination, reporting, recording, planning of all site inspection activities.

EPCM CONSTRUCTION CONTRACTOR shall issue document management and quality records management procedures for Review and approval by EPCM CONTRACTOR and COMPANY.

The special processes (as per Appendix- 3 requirement) require TPI review and/or approval of work procedures, personnel qualifications, or personnel qualification procedures. EPCM CONSTRUCTION CONTRACTOR shall ensure that these special processes are accomplished under controlled conditions as specified in all applicable standards and specifications.

All process parameters shall be identified with acceptance criteria specified and monitored. All Procedures and Personnel Qualification shall be submitted for review and/or approval at least thirty (30) calendar days prior to the start of SERVICES. EPCM CONSTRUCTION CONTRACTOR shall establish KPI with EPCM CONTRACTOR & COMPANY direction and approval of targets for all construction processes and quality measurements.

8.4 EPCM CONSTRUCTION CONTRACTOR QC PROCEDURES

EPCM CONSTRUCTION CONTRACTOR shall, within one (1) month of the contract effective date, submit the list of procedures (EPCM CONSTRUCTION CONTRACTOR Document list) covering all important construction operations as required by the CONTRACT DOCUMENTS and indicating the planned submission date for each document.

The EPCM CONSTRUCTION CONTRACTOR shall submit to the EPCM CONTRACTOR for approval the procedures necessary to perform control activities to the extent that assure full control on the work execution.

Procedures are subject to approval of EPCM CONTRACTOR Quality but not limited to what reference to APPENDIX 3 Control of Construction Process and to COMPANY, where applicable.

All procedures prepared by EPCM CONSTRUCTION CONTRACTOR shall address the following common topics:

- Reference to the applicable ITP
- Reference to the PROJECT specification and international codes and standards
- Scope and purpose of the documents

- Method of procedure qualification / demonstration
- Process flow chart detailing chronological sequence of the full activity
- Inspection methodology and quality requirements
- Acceptance criteria
- Roles and responsibility
- Associated risk and uncertainties

EPCM CONTRACTOR /COMPANY shall have the right to request the submission of further procedures, as he may deem appropriate. In such a case the EPCM CONSTRUCTION CONTRACTOR shall timely submit the required procedures before proceeding with the relevant activities.

The EPCM CONSTRUCTION CONTRACTOR shall submit at least the procedures listed below:

1 – Concrete mix design

The EPCM CONSTRUCTION CONTRACTOR's procedure shall describe the process accordingly with reference to Project concrete supply and construction specification.

2- Fabrication of Piping Spool at Workshop & Inspection Control System for Welding Activities

The procedure shall describe at least:

- The responsibilities involved, the description of shop drawings preparation (including the cutting sheets, material list, spools identification, etc.) and the distribution to the fabrication departments
- The method undertaken by EPCM CONSTRUCTION CONTRACTOR to make sure that only the last revisions of the Approved for Construction (AFC) drawings shall be distributed and used during the welding phases.
- The main operations such as material identification, material traceability, fit-up, welding, visual and dimensional examination, non-destructive test (NDT) operations and post welding heat treatment (PWHT). Importance shall be given to the interface between Fabrication Department and QC Department, the flow of information and the method of communication between Fabrication Department and QC Department.
- The procedure adopted to assure the proper distribution and the correct utilization of approved Welding Procedure Specification (WPS), the responsibilities and system to guarantee that welding parameters and welding electrodes/rods/gas mixtures used during welding activities are in accordance with the applicable WPS.

- The documentation and certification to be prepared and the release of fabricated spools to the field.

And the procedures shall comply with Project Specification for welding and NDT & Specification for shop and field fabrication piping.

3 - Assembly of Piping Spool at Field

The procedure shall describe at least:

- The main operations such as material identification, material traceability, fit-up, welding, visual and dimensional examination, non-destructive test (NDT) operations and post welding heat treatment (PWHT). Importance shall be given to the interface between Fabrication Department and QC Department, the flow of information and the method of communication between Fabrication Department and QC Department.
- The procedure adopted to assure the proper distribution and the correct utilization of approved Welding Procedure Specification (WPS), the responsibilities and system to guarantee that welding parameters and welding electrodes/rods/gas mixtures used during welding activities are in accordance with the applicable WPS.
- The documentation and certification to be prepared and the release of piping test packages for the testing.

And the procedures shall comply with Project Specification for welding and NDT & Specification for shop and field fabrication piping.

4 - Procedure for the Updating of the Welds Status & welding management procedure

The procedure shall describe the method that the EPCM CONSTRUCTION CONTRACTOR shall implement to keep under full control at every stage of the WORKS the welding activities both in terms of production rate achieved and inspections and tests performed.

To this end, the EPCM CONSTRUCTION CONTRACTOR shall record all the relevant data into a computer database, the Welds Status, which shall be updated on a regular basis. Piping welds and structural steel welds shall be kept into separate database.

The computer database operators shall be able to, but not limited to:

- print the status of the Work Progress in terms of number of welded joints and NDT percentage covered for each piping class subdivided by diameter and thickness.
- print the status of the joints welded by each welder.
- Print the quantity and percentage of NDT performed for each welder and the relevant repair rate.

- Verify if all test activities have been satisfactorily completed for each piping class or line or isometric drawing or pipe spool.
- Print all the above information for each isometric drawing.

The procedure shall comply with the project specification for Welding and NDT in addition to AGES-GL-13-001 ADNOC QAQC requirements for contractors in projects.

5 - Procedure for Storage and Handling of Welding Consumables

The procedure shall describe the method that EPCM CONSTRUCTION CONTRACTOR shall implement to guarantee the proper usage of welding consumables (i.e., electrodes, fluxes, wires etc.).

The procedure shall indicate the methods of drying, storing, and handling the welding consumables both in the workshop and the field.

The procedure shall follow the project specification for Welding and NDT in addition to AGES-GL-13-001 ADNOC QAQC requirements for contractors in projects.

6 - Non-destructive tests procedures

EPCM CONSTRUCTION CONTRACTOR shall submit the NDT procedures cover the scope of work and shall be approved and endorsed by EPCM CONSTRUCTION CONTRACTOR NDT LEVEL III specialist and as minimum shall cover below, EPCM CONSTRUCTION CONTRACTOR will be requested to submit special techniques procedures such as PAUT, TOFD with full qualification and calibration, demonstration in case need to be utilized.

All procedures shall be in line with project specification for welding and NDT.

A - Procedure for Visual Examination

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry out the visual examination:

- Of the parts to be welded, before welding,
- Of the completed welds and
- Of the completed items, sub-assemblies, or final assemblies of steel structures, piping and vessels.

The procedure shall specify at least:

- Phases of the examination.
- Inspection and acceptance criteria (for steel structures, piping and vessels).
- Compliance with construction drawings and applicable standards for types, shapes, dimensions, joint categories, and aspect of welds.
- Limits of acceptability of surface defects with the required repair procedure.

- Relevant certification.
- Personnel qualification.
- The procedure shall also include:
- Sketches showing acceptable weld profiles.
- Sketches showing typical imperfections of structural and piping welds.

The procedure shall specify that for the welds or components subject to post weld heat treatment the final visual examination shall be carried out after the heat treatment.

B - Procedure for Liquid Penetrant Examination

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry out the liquid penetrant examination. The procedure shall specify at least:

- Technical restrictions to the applicability of the examination.
- Control of contaminants (if applicable).
- Pre–examination surface preparation.
- Liquid penetrant products and their grouping (i.e., the combination of cleaner, penetrant, remover and developer by type or supplier).
- Examination procedure.
- Interpretation of the results.
- Acceptance and rejection criteria.
- Defect removal procedure.
- Restoring after the examination.
- Certification
- Personnel qualification.

The procedure shall specify that for the welds or the components subject to post weld heat treatment the final examination shall be carried out after the heat treatment.

C - Procedure for Magnetic Particle Examination

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry out the magnetic particle examination. The procedure shall specify at least:

- Surface preparation.
- Test equipment and relevant calibration.
- Examination mediums.

- Magnetising methods.
- Examination procedure.
- Magnetising force adequacy.
- Evaluation of the results.
- Acceptance/rejection criteria.
- Defect removal procedure.
- Demagnetising (if necessary).
- Restoring after the examination.
- Certification.
- Personnel qualification.

The procedure shall specify that for welds or components subject to post weld heat treatment the final examination shall be carried out after heat treatment.

D - Procedure for Radiographic Examination

The procedure shall describe how the SUBCONTRACTOR shall carry out the radiographic examination of steel butt-weld joints by X-ray and/or Gamma-ray in accordance with SUBCONTRACT requirements. The procedure shall specify at least:

- Material type and thickness range, surface condition.
- Radiation sources, isotope or X-ray tube used.
- Radiographic films brand and designation.
- Screens used.
- Required sensitivity.
- Selection of penetrometer.
- Positioning of penetrometer.
- Number of penetrometers.
- Radiographic technique.
- Radiographic sensitivity.
- Acceptable film density.
- Maximum source size.
- Minimum source–to–object distance, maximum distance from source side of object to the film.

- Radiograph identification system (see note *' below).
- Acceptance standard.
- Defect removal.
- Certification.
- Personnel qualification.

The procedure shall specify that for welds or components subject to post weld heat treatment the final examination shall be carried out after heat treatment.

Note (*): an identification system shall be used to ensure a permanent record of the welds to be radiographed and the films.

As a minimum the following identification shall appear on the radiographic image (lead markers must be placed on the part to be examined). Additional information may be required by the CONTRACTOR or the COMPANY:

-drawing number.

-weld number (the label "R" shall be added in case of repair).

-film sequence (metric tape with lead digits).

-date of execution.

-In the envelope of the set of films related to each weld, the above-mentioned identifications must be repeated, and the following must be added:

-SUBCONTRACTOR's logo or name.

-welder identification number (stamp).

-job number.

EPCM CONSTRUCTION CONTRACTOR shall prepare RT film Digitization procedure for Welds as stated in the Project specification and the same shall be implemented for all EPCM CONSTRUCTION CONTRACTOR &their Subcontractors, fabrication yards, Vendors & Piping/Pipeline welds. RT Digitization is applicable for both Piping (shop and site Welds), Pipeline and other equipment / packages as requested in Project specification. RT film digital Image shall meet requirements of ASME V Article 2 Mandatory Appendix VI.

E- Procedure for Ultrasonic Examination

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry out the ultrasonic examination on welded joints and base materials when this type of inspection is required by applicable documents. The procedure shall specify at least:

- ultrasonic apparatus, probes, and the coupling medium.
- surface preparation.

- calibration of test equipment:
- Angle beam calibration (check of probe angle, calibration of range scale, and amplitude calibration).
- Straight beam calibration (calibration of range scale and amplitude calibration).
- determination of surface compensation.
- definition of scanning techniques related to joint typology.
- inspection method.
- evaluation of examination results.
- acceptance standards.
- certification.
- personnel qualification.

The procedure shall specify that for welds or components subject to post weld heat treatment the final examination shall be carried out after heat treatment.

7- Procedure for Dimensional Control

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry out the dimensional control during the fabrication of structural members and piping spools at workshop and during the erection of structures, piping, and equipment at field.

The procedure shall specify at least:

- survey equipment and limits of application (steel tapes, ultrasonic thickness gauges, optical levels, theodolites, steel square, steel rules etc.).
- phases of check (e.g., after cutting and beveling, after fit-up, after welding of prefabricated items, after fabrication completion etc.).
- critical dimensions to be checked.
- certification.
- The dimensional control certificates shall report at least:
- actual dimension.
- nominal dimension tolerance.
- acceptance or rejection

8- Procedure for Post Weld Heat Treatment of Carbon and Low Alloy Steel Piping

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry out the heat treatment. The procedure shall specify at least:

- Type of materials and thickness limitation.

- heating methods and equipment.
- preparation.
- furnace heating (if applicable).
- local heat treatment.
- thermocouples type and location.
- heating cycle (heating rate/holding time and temperature/cooling rate).
- PWHT chart, time/temperature recorder calibration.
- inspection and documentation.

9 - Erection of Steel Structure

EPCM CONSTRUCTION CONTRACTOR's procedure shall describe at least:

- Alignment check
- Bolting and torquing

10 - Electrical and Instrumentation Works

The procedures shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry the electrical and/or instrumentation works, as applicable under the SUBCONTRACT. The document shall specify at least:

- identification and check of equipment, instruments and bulk material before installation.
- care of and preservation of equipment/instruments after their withdrawal from warehouse.
- equipment/instruments installation general operations (e.g.: location, levelling and alignment of switchgear/switchboard panels, Power centre; alignment and bolts tightening of bus bars, etc.).
- cables and accessories installation (e.g.: sealing of cuts on cable drums and on not terminated cables, laying on cable racks and routing, cables dressing and fastening, bending radius, cable and cores identification, cables terminations, etc.).
- updating of cable list tables with the as built pulling length.
- cable racks/conduits installation operations (e.g.: interference-hazard check, electrical continuity, spacing of supports and bolting etc.).
- prefabrication and painting/galvanizing of electrical and instrument supports.
- inspection and tests (check of the correspondence of the electrical & instrumentation installations to the standards and project documents, cables continuity and insulation, winding test, oil dielectric test, etc.).

- testing equipment and relevant calibration, field laboratory for instrument calibration.
- instrumentation piping and tubing installation and pressure test
- loop tests and functional tests.
- safety requirements.
- personnel qualification.

11 - Surface Preparation and Painting Control Procedure

The procedure shall describe how the EPCM CONSTRUCTION CONTRACTOR shall carry the surface preparation, paint inspection and testing for structures, piping, equipment, incompliance Project specification for painting and coating.

The document shall specify at least:

- preliminary activities.
- surface preparation.
- equipment and materials for sand blasting.
- sand-blasting method and inspection (weather condition monitoring, cleaning grade and roughness control).
- protection of the areas to be welded.
- coating application.
- equipment for primer and paint applications.
- inspection of coating (weather condition, wet and dry film thickness measurements, adhesion tests).
- measures to avoid over spray on adjacent components.
- inspection and testing devices and relevant calibration.
- remedial work and touch-up.
- certification.
- safety requirements.
- sandblaster and painters' qualification

EPCM CONSTRUCTION CONTRACTOR QC procedure shall be approved by EPCM CONTRACTOR and COMPANY.

Before starting any coating system application, EPCM CONSTRUCTION CONTRACTOR shall invite EPCM CONTRACTOR and/or COMPANY to mock-up sections for validation of the painting systems and qualification of applicators.

This will also apply to eventual coating applications for fireproofing purposes.

12- Pipeline procedures

Shall cover pipeline stacking, welding, laying, coating and testing, drying, preservation. Tie ins execution and testing, Hot tapping procedures, welding and testing.

13 - GRE piping

For GRE piping, VENDOR's instructions/specifications shall be used and all qualification for laminators and bonders shall be approved by EPCM CONTRACTOR/COMPANY.

14 - HDPE piping

EPCM CONSTRUCTION CONTRACTOR shall prepare a procedure for Thermal contact electro fusion coupling.

EPCM CONSTRUCTION CONTRACTOR shall also provide training of Bonder or Bonding operator/Fusing operator and records, Bonder or Bonding Operator performance qualification (BPQ) and, in addition:

1. Bonding Procedure Specification (BPS)
2. Bonding Procedure Qualification Record (PQR/WPAR).
3. Joining Book: Joining Map & BPS with supporting PQR/WPAR's.

8.5 METHOD OF STATEMENTS

EPCM CONSTRUCTION CONTRACTOR shall, within one (1) month of the award date, submit the list of method statements covering all important construction operations as required by the SUBCONTRACT SOW and indicating the planned submission date for each document.

Method Statement is the document prepared by the EPCM CONSTRUCTION CONTRACTOR describing the method that EPCM CONSTRUCTION CONTRACTOR intends to put in practice to carry out a particular part of the WORKS.

The EPCM CONSTRUCTION CONTRACTOR shall submit the Method Statements to the EPCM CONTRACTOR for approval to the extent that the same are applicable to the WORKS.

Method statements are subject to approval of EPCM CONTRACTOR Site Construction Group.

All Method Statements prepared by the EPCM CONSTRUCTION CONTRACTOR shall address the following common topics:

- Scope
- Reference Documents
- Definitions

- Responsibilities

EPCM CONTRACTOR shall have the right to request the submission of further method statements, as he may deem appropriate. In such a case the EPCM CONSTRUCTION CONTRACTOR shall timely submit the required method statements before proceeding with the relevant activities.

The EPCM CONSTRUCTION CONTRACTOR shall submit at least the statements listed below:

1) Site preparation

The method statement shall describe:

-backfilling of site area: methods, equipment, and testing.

-Acceptance of activities.

-survey grid.

2) Piling

The method statement shall describe:

-boring and pile bore logs / casing / reinforcing steel / casting.

-pile testing.

-concrete pile destructive trial tests.

-bearing load capacity spot test.

-ultrasonic integrity test.

-iron piling / prefabricated piles / others.

-safety requirements.

3) Foundations

The method statement shall describe:

-digging of foundations / lowering of water table.

-formworks / reinforcing steel / concrete casting and curing / waterproof and/or coatings.

-backfilling around foundation and buildings.

-dimensional control.

-release process of the foundation as those general items their release considered commissioned.

-safety requirements

4)Buildings

The method statement shall describe:

-wall erection.

-reinforced concrete / prefabricate floor slabs.

-roofs and waterproofing.

-floors: tiled floors / floating floors / false floors, various.

-earthing system.

-insulation: thermal / acoustical / electrical.

-ceilings / false ceilings.

-H.V.A.C.

-windows and door frames.

-telephones lines, cabling, etc.

-dimensional control.

-safety requirements

5) Road Works

The method statement shall describe:

-road sub grade

-asphalts.

-safety requirements

6)External Paving

The method statement shall describe:

-gravel, concrete, asphalt or tiles

7) Underground

The method statement shall describe:

-Sewage net, pits, etc.

-safety requirements

Other method statements dedicated to construction of particular importance (e.g., Concrete cementitious fireproofing, Water Pump Houses, Control Rooms, Tank Foundations, Drop Spillways, etc.)

8) Installation of Equipment

-The method statement shall describe how the SUBCONTRACTOR carry out the installation of the equipment. The document shall specify at least:

-identification and check of equipment, loose components, and bulk material before installation.

-care of and preservation of equipment after withdrawal from CONTRACTOR's warehouse.

-equipment installation general operations (e.g.: location, levelling and alignment of equipment, alignment and bolts tightening, etc.).

-accessories installation.

-inspection and tests

-documentation.

-safety requirements.

-personnel qualification

Other method statements dedicated to works of particular importance (e.g., Heavy Liftings, Large Pumps Installation, etc.)

9) Insulation Works

The method statement shall describe how the SUBCONTRACTOR shall carry the insulation works.

The document shall specify at least:

fabrication of jacketing.

•application of insulation materials.

•application of vapor barrier (if required).

•installation of jacketing

•remedial works.

•certification.

•safety requirements

•personnel qualification.

10) Pipeline execution method statements including surveying, ROW, trenching, road crossing, laying, compaction, rout marking, etc.

9.0 DOCUMENT CONTROL

9.1 TECHNICAL DOCUMENTS

EPCM CONSTRUCTION CONTRACTOR shall implement an Electronic Document Management System (EDMS) and shall submit to EPCM CONTRACTOR & COMPANY all drawings, documents, transmittals and quality records, etc. from initial construction stage to final Document Handover. This shall be the main system utilized for all EPCM CONSTRUCTION CONTRACTOR and COMPANY Document Control requirements and EPCM CONTRACTOR & COMPANY shall have free online access. EPCM CONSTRUCTION CONTRACTOR shall provide its procedures for this area, together with its standard organization. Master register shall be maintained by the document controller for all the project documents

EPCM CONSTRUCTION CONTRACTOR Technical documentation at Site shall be managed according to robust procedure for document management at site issued by the EPCM CONSTRUCTION CONTRACTOR and approved by EPCM CONTRACTOR /COMPANY.

The EPCM CONSTRUCTION CONTRACTOR shall properly manage the technical documentation received from the EPCM CONTRACTOR, as well as deliverables produced under EPCM CONSTRUCTION CONTRACTOR scope.

To this end, the EPCM CONSTRUCTION CONTRACTOR shall orderly record all the aforesaid documents in a dedicated register, which shall be regularly updated and submitted to EPCM CONTRACTOR/COMPANY.

Records shall be protected from destruction by fire, flooding, tornadoes, insects, and rodents; from deterioration by extremes in temperature and humidity; and against theft or corruption.

EPCM CONTRACTOR shall monitor EPCM CONSTRUCTION CONTRACTOR document management during periodical audits.

EPCM CONSTRUCTION CONTRACTOR shall develop the sub system and system index for all discipline works and to submit sub-index for each file and volume compilation. all documents and records shall be hyperlinked with document or report number. This shall include quality records, material test certificates and heat numbers used for construction, all tagged items etc.

EPCM CONSTRUCTION CONTRACTOR shall demonstrate a power point presentation to EPCM CONTRACTOR/COMPANY for better clarity and understanding of electronic filing, documentation and handover integrating all discipline works.

9.2 QUALITY RECORDS

EPCM CONSTRUCTION CONTRACTOR shall keep and collect all original tests reports (civil, concrete, compaction, testing, surveying, welding, NDT, PWHT, RT, VT, UT, MT, PMI, etc) that will be submitted to EPCMCONTRACTOR /COMPANY for endorsement during execution and at the end of their site activities refer to requirement in section 7 above.

EPCM CONSTRUCTION CONTRACTOR shall implement a robust Inspection notification and quality records maintaining, and shall be digitalized through EDMS system, all inspection records shall be traceable its related line numbers, systems, Tag numbers, drawing etc, & recorded in electronic/digital document management system.

EPCM CONSTRUCTION CONTRACTOR shall issue document management and quality records management procedures for Review and approval by EPCM CONTRACTOR and COMPANY.

10.0 CRITICAL ACTIVITIES

10.1 Welding activities

Welding activities are regulated by Project welding specifications & Construction procedures included in the Scope of Work issued to the EPCM CONSTRUCTION CONTRACTOR.

EPCM CONSTRUCTION CONTRACTOR shall demonstrate that control of operation involving welding conforms as minimum to ISO 3834-1 ‘’ quality requirements of fusion welding of metallic material’’ to the level of detailed Comprehensive Quality Requirements in ISO 3834-2 latest editions

EPCM CONSTRUCTION CONTRACTOR shall submit to EPCM CONTRACTOR/COMPANY for approval all necessary documents, well in advance before starting any welding activities. EPCM CONSTRUCTION CONTRACTOR shall identify all welding procedure and submit Preliminary Welding Procedure Specification (PWPS) for EPCM CONTRACTOR and COMPANY approval. PWPS format shall be agreed and endorsed by EPCM CONTRACTOR /COMPANY. No qualification shall be performed without EPCM CONTRACTOR/COMPANY approval of PWPS.

The Welding Procedures and Welder Qualification to be done specific to the project. Previous Qualified Welding procedures not acceptable

Approved WPS shall be available at the moment of related welding activities commencement. WPS shall clearly identify the PROJECT Material Class number.

Welders shall be qualified specific for the project both for Shop and Site welding in construction scope. No Previous Qualifications are accepted

All dissimilar metal welded joints should be avoided whenever practical and are not allowed in sour service PIPING CLASSES as defined in Specification

PWHT requirements for piping in sour wet H2S and other severe service shall be in accordance with AGES-SP-07-003.

PWHT for Pipe pre-fabrication and site welding to be carried out by Independent third-party SUBCONTRACTOR qualified in ADNOC Commercial Directory (AVL).

PWHT of piping systems for Sour Service shall be as specified below: Process Piping and Pipe Fittings:

In a sour service / wet H2S service environment, PWHT is mandatory for the following cases

a) Double sided welding, i.e., flange girth welds.

b) Internal weld repair.

c) Internal weld build-up for dissimilar thickness joints to overcome misalignment. The applicable industry specs must be followed for PWHT requirement for any service not listed above. Other process conditions may also require PWHT, as determined during the project design. Code exemptions for PWHT are not permitted if PWHT is specified for process conditions.

For piping components, PWHT is exempted if:

a) Weld thickness is more than 5mm and ≤ 19 mm.

b) Weld is not made from inside.

c) Hardness values are in conformance to the requirement of NACE MR0175 / ISO 15156 - 2.

d) Weldment is multi-pass welds.

e) PWHT is not specified by design code.

<u>PWHT is mandatory:</u>

a) If not meeting any of the above requirements from a to e, however since PWHT is an essential variable, a qualified weld procedure based on PWHT shall be required to be approved by COMPANY.

b) Where production hardness values are not met; or where,

c) PWHT is specified by Licensor.

PWHT for the non-Sour service, AGES shall be followed.

Weld deposits found to exceed the maximum hardness criterion are unacceptable and shall be reported to the EPCM CONTRACTOR & COMPANY. Unless accepted by the EPCM CONTRACTOR & COMPANY, hard welds shall be both removed and rewelded, or heat treated to reduce the hardness to an acceptable value. The specific approach to be used to correct the high-hardness condition shall be subject to the EPCM CONTRACTOR approval before any corrective action is taken. Regardless of the method of corrective action taken,

the weld deposits shall be retested to ensure that the corrective action has resulted in acceptable hardness values.

Temperature uniformity Survey shall be carried out as per applicable ASTM standard by placing the t/c in all the zones for temporary constructed furnace and 100 % production hardness to be carried out.

Production hardness testing for piping in sour and HIC services shall be in accordance with AGES-SP-07-003.

All welding procedure qualification tests and welder performance qualification tests shall be subject to witnessing and approval by EPCM CONTRACTOR and COMPANY TPI and endorsed by EPCM CONSTRUCTION CONTRACTOR TPI.

Filler metal certification shall be in accordance with ASME SFA-5.01 and PROJECT specification for welding and NDT.EPCM CONSTRUCTION CONTRACTOR shall submit filler metals certificates for review before starting of welding TO EPCM CONTRACTOR/ COMPANY, who has the right to request certain test to verify filler metal compliance with applicable PROJECT requirements.

EPCM CONSTRUCTION CONTRACTOR shall keep the information on welds updated by using a dedicated Welding Database.

EPCM CONSTRUCTION CONTRACTOR shall consider that repair rate / percentage shall be calculated based on the number of tested joints by RT, UT, MT or DPT (as applicable), for pipelines & piping, while to be calculated by metric length of tested joints for storage tanks, pressure vessels, steel structures and other plate work considering the following:

Repair rate / percentage shall not exceed 3% of the total number of welded joints or 0.5% of the total welded length in a week as detailed above.

In case the weekly repair rate / percentage exceeds above limit, a full detailed investigation shall be carried out by the EPCM CONSTRUCTION CONTRACTOR to detect the root cause of the defect, eliminate it, and implement corrective measures to prevent recurrence and reduce the repair rate / percentage to within maximum limit.

When weekly repair rate / percentage exceeds 5% of the total number of welded joints or 1% of the total welded length, EPCM CONSTRUCTION CONTRACTOR shall be officially warned by EPCM and COMPANY and WORK shall be suspended until satisfactory completion of the investigation and root cause analysis associated with the courses of rectification, corrective, and correction actions which shall be accepted and approved by COMPANY.

Weekly as well as overall cumulative weld repair rate / percentage shall be reported on weekly basis through the periodic Quality reports & EPCM CONSTRUCTION CONTRACTOR to submit Trend analysis for weld repairs arise from any NDE examination and submit to COMPANY on weekly basis. Monthly reports shall include overall cumulative weld repair rate.

EPCM CONTRACTOR shall continuously monitor the weld /welder repair rates and shall be proactive in the elimination of the potential causes that might lead to critical limits.

EPCM CONSTRUCTION CONTRACTOR shall track weld production, welder performance, productivity of each welder and welding groups, weld reject rates, and NDT status including status of weld repairs, and any required penalty tracer shots on a daily and weekly basis and submit results to EPCM each week (seven calendar days).

EPCM CONSTRUCTION CONTRACTOR shall review and track PROJECT errors & failures such as welding & welder repair rates daily. Welders' repair rate / percentage shall be monitored on a weekly basis and shall not exceed 3% of the total welders' weekly production; also weld repair rates and on-spot welders' performance shall be evaluated on a weekly basis.

Welders Personnel Qualification Certificates, welder qualification reports, plans and procedure, to be part of the final dossier.

If any welder or welding operator who consistently fails to meet the standard shall be required to undergo re-training, re-testing or shall be removed from the job completely.

EPCM CONSTRUCTION CONTRACTOR shall maintain WPS Log for all WPSs utilized applicability and ranges, etc. for piping and structure.

10.2 NON-DESTRUCTIVE TESTING (NDT)

EPCM CONTRACTOR shall review/witness and accept results of all NDT performed by EPCM CONSTRUCTION CONTRACTOR and submit test results to COMPANY.

EPCM CONSTRUCTION CONTRACTOR shall be fully responsible for the performance and control of its construction process including their lower tier sub-contractor and their sub-supplier NDT activities.

All Non-Destructive Testing (NDE) shall be carried out by independent Sub-Contractor Qualified in the relevant product group ADNOC Commercial Directory.

The NDE company responsible for the final examination of welds shall maintain a quality system in accordance with ISO / IEC 17020, or equivalent as agreed with the COMPANY, and be accredited by a nationally recognized body.

All NDT procedures shall be prepared by qualified and certified ASNT ACCP NDT Level III or equivalent certification approved by EPCM and COMPANY.

PMI testing shall be performed when required by and in accordance with AGES-GL-13-002.

Hardness testing of production welds shall be carried out at five locations including base metal 1, Base metal 2, center of the weld. Close to HAZ at both sides of the weld in the same plane and at 180 degree apart in the same weld joint. The readings should be reported in the hardness report for evaluation and acceptance.

Temperature uniformity Survey shall be carried out as per applicable ASTM standard by

placing the t/c in all the zones for temporary constructed furnace and 100 % production hardness to be carried out.

Production hardness testing for piping in sour and HIC services shall be in accordance with AGES-SP-07-003.

When there is a requirement for NDT examination in percentage wise of the total number of welds the following requirements shall apply:

- Welds shall be randomly and independently selected throughout the fabrication / construction period and fully examined.
- The random selection shall be equally distributed overall the welders, all type of welds, all welding procedures, all pipe sizes / thicknesses, and all welding position.
- Where reduced percentage of examination is permitted and the resulting test shows defects, then two additional samples of the same kind welded by the same welder shall be given the same type of examination.
- If any of the items found defective, then two further samples of the same kind shall be examined for each defective item found by that sampling.

This shall continue until fully examined and repaired or replaced as necessary to meet the specification requirements.

If the weld reject ratio per welder or welding operator exceeds 5% RT shall be increased to 10% and stay at 10% until the reject rate per welder or welding operator drops below 5% and remains below 5% for another 14-consecutive working days.

EPCM CONSTRUCTION CONTRACTOR shall submit NDT MAP covering the piping class, Rating, Type of Joint, Material, service type, fluid type and % of NDT including % Hardness, Ferrite testing, PMI etc.

EPCM CONSTRUCTION CONTRACTOR shall prepare a procedure specifying minimum requirements for the qualification of NDT, for types of equipment, and for types of NDT used and subject the procedure for review and approval of the COMPANY.

SL. No	Pipe Class	Rating	Facing	Material	Fluid Service	Lethal	PWHT (Note-11 Sour, Note-13 Non-Sour)	Sour / Non-sour	Design Conditions (Note-15)		Service	Maximum Size (NPS)	Extent of NDE (Note-2)					Remarks
									Design (Max) Press., barg	Design (Min/Max) Temp. Range, oC			RT% (Note 3,4,5,6,7)	MT% (Note 8)	LP%	PMI%	Hardness% (Note 24)	

EPCM CONSTRUCTION CONTRACTOR shall update the table if any additional Piping classes / Rating added in the PMS. Updated document shall be submitted to for approval.

1. All the Welds shall be 100% visually inspected and recorded.

2. NDE requirement has been provided based on the material and pipe class and will be further updated based on engineering development by process.

3. The radiography requirements are applicable to butt-welds, branch connection welds suitable for radiography. If branch connection cannot be radiographed, then MT or Liquid penetrant examination shall be performed at the same time, percentage as required for radiography for that service.

The RT films shall be electronically archived through radiographic digitalization for all fabrication and site construction activities which shall be in acceptable quality level to the extent of COMPANY satisfaction and shall follow international norms and good practices. A dedicated PROJECT procedure for Digitization of Radiographic Films shall be made available for SUBCONTRACTOR specific Scope of Work.

Storage of RT films during construction shall be in accordance with international codes, standards and manufacturer recommendation.

For Socket welds, the first 10 production welds shall be examined by radiography and on subsequent welds (during construction) 5% or 10% radiography as mentioned in above table shall be carried out on random basis. In Lieu of RT, the EPCM CONSTRUCTION CONTRACTOR may propose other technologies like Gap-O-Let, water-soluble spacer rings with 100% socket weld joints etc. in order to ensure the gap as specified in ASME B31.3 for Company Approval.

4. PAUT examination may be applied if the configuration precludes radiography.

5. The indicated percentage of welds to be radiographed shall represent welds made by each welder’s and welding operator’s work, including the small-bore nipples.

6. Small-bore nipples to be included (Socket welds for Non sour service and butt welded for Sour service).

7. For X-ray-fine grain may be used, e.g., Agfa D7, Kodak Industries AX. For Gamma-ray ultra-fine grain film shall be used, e.g., Agfa D4, Kodak Industries MX.

8. MT - Magnetic Particle Inspection.

In addition to buttwelds, Fillet welds shall be subject to MT (Magnetic Particle Examination) / Liquid penetrant examination at the same percentage as for the radiography of butt welds.

For jacketed piping, 10% of the circumferential and 100% of the longitudinal welds shall be examined by radiography.

9. The buried pipe shall be externally protected in accordance with AGES-SP-07-002 “External Pipeline Coatings Specification”, and AGES-SP-07-001 “Cathodic Protection Specification”.

Records shall be protected from destruction by fire, flooding, tornadoes, insects, and rodents, from deterioration by extremes in temperature and humidity and against theft or corruption.

Retention of all RT films shall be the responsibility of EPCM construction CONTRACTOR for a period of not less than 3 years after Mechanical completion.

Records shall be protected from destruction by fire, flooding, tornadoes, insects, and rodents, from deterioration by extremes in temperature and humidity and against theft or corruption.

NDT/NDE Personnel Qualification Certificates shall be in line with COMPANY specification AGES-GL-13-001,

NDT/NDE Personnel Qualification Certificates, NDE reports, NDE procedures, to be part of the final dossier.

Advanced NDT Methods:

PAUT shall be carried out by ADNOC approved Contractor for specialized NDE. The Technician carrying out PAUT shall have ISO 9712 level-II and have training on the relevant equipment that will be used. The procedure and scan plan shall be approved by PCN Level -III / ISO 9712 Level -III in PAUT. PAUT shall be witnessed by COMPANY approved by EPCM CONTRACTOR TPI having certification in PCN Level -II / ISO 9712 Level -II in PAUT with 5 years of experience as a minimum.

First 20 joints of PAUT spread across all thickness and dia for each SUBCONTRACTOR carrying PAUT shall be verified by RT and comparison report submitted to COMPANY. This shall be agreed with COMPANY prior to start of PAUT.

PAUT mandatory for:

a) Thickness ≥ 25 mm

b) Diameter ≥ 20” and thickness ≥ 12 mm

In addition, all joints in Brown field and joints during shutdowns shall be carried out with PAUT. Alternate NDE method are subjected to Company’s approval.

EPCM CONTRACTOR to submit a NDE plan for PAUT including the thickness range, diameters, calibration block and Validation (Demonstration) block details for the complete range of pipe system planned with PAUT

10.3 PICKLING AND PASSIVATION

Pickling and passivation at Site shall be carried out only on weld joint areas, whereas others Corrosion Resistant Alloy surfaces shall be pickled and passivated at shop (when applicable and required).

10.4 PAINTING AND COATING

All painting and coating procedures shall be prepared by qualified EPCM CONSTRUCTION CONTRACTOR Personnel and certified NACE Level III or equivalent certification approved by EPCM CONTRACTOR and COMPANY.

Painting/Coating personnel qualification certificates, plans, reports, to be part of the final

dossier.

supervisors and inspectors' deployment plans shall be in Line with AGES-GL-13-001 AND appendix 01

EPCM CONTRACTOR Pre award audits shall ensure digital programmed software & applications for manufacturing, construction, planning, execution and their controls logic gates shall be verified.

10.5 FLANGE MANAGEMENT

EPCM CONSTRUCTION CONTRACTOR shall develop a detailed flange management procedure for implementation during all phases of work to ensure Maximum Integrity of Flange Joints. EPCM CONSTRUCTION CONTRACTOR shall submit this procedure at least one month prior to start of the Construction activities for review and approval.

EPCM CONSTRUCTION CONTRACTOR shall maintain a comprehensive flange integrity management software to plan, monitor, and control and provide traceability of flange joints including real time status of the Flanged Joints. EPCM CONSTRUCTION CONTRACTOR shall demonstrate the software to EPCM CONTRACTOR & COMPANYs' satisfaction. EPCM CONSTRUCTION CONTRACTOR shall provide access to COMPANY for this Software.

As a minimum the following shall be addressed in the Flange management system.

a. Practices to be followed during the Life cycle of the Project, starting from protection/tagging of flanges at vendor works through Fabrication shops, installation of Flanges at site and till Commissioning.

b. Competency Requirements:

1.Fitters & Supervisors with prior experience in Flange jointing shall only be employed.

2.All fitters shall have passed and certified through a Training Institute.

3.EPCM construction CONTRACTOR shall develop a detailed training program including classroom session for all the fitter and supervisors employed for Jointing.

4.The Competency of the fitters employed shall be demonstrated to COMPANY satisfaction prior to deployment.

5.Each Fitter shall be given a specific card addressing the type of jointing (manual, Torquing, tensioning etc...).

c. Acceptance criteria for flange faces prior to tightening

d. Installation of different types of gaskets including insulating kits.

e. Methods of tightening the bolts – Tensioning, Torquing etc. with sequencing and torque values.

f. Each flange shall be uniquely identified in the Isometrics. Each Flange joint shall be tagged

to know the status of the joint.

g. EPCM CONTRACTOR shall develop and maintain a database which address the following:

1. Traceability to the Line number & Isometric.

2.Flange, Bolt & Nut, Gasket details.

3.Insulation requirements.

4.Type of Tightening and Torque data.

The EPCM CONTRACTORs construction Quality team shall sign off for each joint and further signed off again by Pre-Commission team.

h. Storage and Preservation of Flanges, Fasteners, Kits, Gaskets during the life cycle of the Project.

i. Format of Construction and Pre-Commissioning Checklist.

Flange Torquing, tensioning, Gasket verification and final Construction sign off in the Tag shall be:

a. Hold points for EPCM CONSTRUCTION CONTRACTOR.

b. EPC CONTRACTOR /COMPANY may have the option of witnessing the above activities. In addition, HC equipment nozzles-piping joints, Battery Limit connections, Pipeline station piping connections may be a Hold point for COMPANY.

c. Each flange after tightening shall have a tag which shall contain System details, Line no, Isometric No, Joint No and date of completion of tightening. The rear portion of the Tag shall contain minor check list of the following as minimum - Cleanliness and alignment of the faces, type, size and material of the Gasket, correctness of bolt / stud length and material, Type of fastening (Torquing, tensioning and Manual) with torque value. Further the tag shall have perforated multiple sections for sign off and after assembly completion, after tightening completing and after final acceptance. The perforated sheets shall be traced to Joint no. The perforated tags for each flange joint shall be part of the Hydrotest pack.

The above shall be addressed in the Inspection and Test plan (ITP).

10.6 HYDROSTATIC TEST PACKAGE REQUIREMENTS:

EPCM CONSTRUCTION CONTRACTOR shall submit Hydrostatic Test Packages prior to conducting hydrostatic tests of piping, pipelines and equipment's for review and approval. The hydrostatic test package shall include, but not limited to, the following:

1)Hydrostatic Test Procedure and test report

2)Hydrostatic Test Diagram which shall include the limits of piping (including test manifold) and equipment included in each hydrostatic test, test pressures, test temperature, test fluid,

line flushing requirements, and safety precautions.

3)Hydrostatic test punch list

4)Relief Valve datasheet and calibration records.

5)Pressures gages and recorders calibration records.

6) Safety Instruction Sheet and Pressure Test Report Form.

7)Copies of P&ID and isometric drawings of the piping system to be hydrostatically tested.

8)Piping Spools control sheet with NDT extent and results.

9)Inspection & Test Plan signed off.

10)Pre-hydrostatic test Check List.

11)Re-instatement procedure and Check List.

12)Bolting and Gasket Control Sheet.

13)Temporary blinds list for hydrostatic test along with standard drawing certified by Designer.

14)Water quality certificate

15)Line list

16)Technical query or site instruction applicable for modification if any

17)Special Pipe supports drawings

10.7 MECHANICAL COMPLETION (INCLUDING PRE-COMMISSIONING)

This activity shall not start until all QC checks have been completed and signed off.

EPCM CONSTRUCTION CONTRACTOR side by side with EPC CONTRACTOR shall put in place procedures which plan and identify all SYSTEM tests required and ensures that each activity is formally verified by signed documentation/certification.

All SYSTEMS shall be fully punched out by EPCM CONTRACTOR before offering for MECHANICAL COMPLETION. COMPANY sign-off on all appropriate documents shall be made before any SYSTEM/ equipment is energized for testing. Status of all testing shall be maintained on an EPCM CONTRACTOR's proven electronic database throughout the MECHANICAL COMPLETION/pre-commissioning activities.

The description will address the essential characteristics used for tracking outstanding construction work items necessary for establishing completion of facilities and components for turnover to the COMMISSIONING and Start-up organization. EPCM CONSTRUCTION CONTRACTOR with EPCM CONTRACTOR shall identify process and non-process systems and sub systems for mechanical completion and hand over of the project. Based on systems

and sub systems definition for piping, piping circuits shall be identified to complete the systems and sub systems.

The piping circuits are generally called hydro test or pressure test packages and shall be completed to cover the entire P&ID. EPCM CONSTRUCTION CONTRACTOR shall assign separate task force and inspection team to perform daily walk down, punching, punch list preparation, test pack documentation and completion of pressure test. Test pack database shall be established and developed to track and manage the test packs and system completion. All Category ‘A” punch points shall be closed before submission of test packages to EPCM CONTRACTOR & COMPANY. Category ‘B” punch points shall be closed during pre-commissioning. Category ‘C” punch points can be closed prior or during commissioning.

11.0 EPCM CONSTRUCTION CONTRACTOR MATERIAL

11.1 CONSTRUCTION INSPECTION AND TESTING ACTIVITIES

No material provided by the CONSTRUCTION CONTRACTOR shall be incorporated into the works without the prior approval of the EPCM and/or COMPANY.

Construction contractor shall submit site material approval request procedure for EPCM/COMPANY approval and shall comply to project standards and quality requirements.

11.2 EPCM CONSTRUCTION CONTRACTOR MATERIAL MANAGEMENT

The management of all materials supplied by the EPCM CONSTRUCTION CONTRACTOR (If required) or free issue shall be under the responsibility of the EPCM CONSTRUCTION CONTRACTOR.

The project material or EPCM CONSTRUCTION CONTRACTOR’s materials that cannot be left outdoors exposed to environmental agents shall be stored in closed warehouses provided by the EPCM CONSTRUCTION CONTRACTOR.

When EPCM CONSTRUCTION CONTRACTOR is requested to provide material to be incorporated into the works, shall submit the Material Approval Requests (MAR) to EPCM CONTRACTOR for approval. Material shall be supplied only from COMPANY approved vendor list, unless differently approved via Stock material approval request.

All material shall meet requirements of design data requisitions, specification, and relevant local code & standard.

To ensure the fulfilment of the aforesaid provision, the CC shall comply with the obligations set forth in the following sub-paragraphs.

EPCM CONTRACTOR shall control EPCM CONSTRUCTION CONTRACTOR supplied materials and fabricated assemblies that do not conform to requirements. EPCM CONSTRUCTION CONTRACTOR shall segregate all nonconforming material to a clearly quarantine rejection area under their responsibility.

11.3 APPROVAL OF LOW TIER SUBCONTRACTORS/VENDOR

EPCM CONSTRUCTION CONTRACTOR Lower tier SUBCONTRACTS shall be placed only with approved SUBCONTRACTORs from COMPANY ACL (Approved contractors list) & AVL (Approved vendor list).

Prior to purchasing any material, the EPCM CONSTRUCTION CONTRACTOR shall submit to the EPCM CONTRACTOR for approval, the list of all proposed low tier vendors/ SUBCONTRACTORS.

The list shall indicate for each category of materials the name of the proposed vendor, the name of the manufacturer and shall enclose the relevant catalogues.

EPCM CONSTRUCTION CONTRACTOR shall utilize only approved laboratories for NDT, PWHT, and other special processes and material testing, civil material testing, concrete testing etc.

Any new supplier required during the execution of the work shall be from COMPANY approved Subcontractor's List.

11.4 CERTIFICATION OF MATERIALS

Whereas the Site Material Approval Request procedure and Certification of material are valid for all EPCM CONSTRUCTION CONTRACTOR Material, for quarry material, the source of material shall be managed as here described.

The EPCM CONSTRUCTION CONTRACTOR shall document the origin of the materials through the submission of the relevant certificates issued by the supplier. The EPCM CONTRACTOR /COMPANY shall have the right to reject any materials without the proper certification.

The certificates of the materials shall be included in the final documentation as per SUBCONTRACT requirements.

Unless otherwise instructed by EPCM CONTRACTOR, the original true copies stamp of the certificates shall be kept by the EPCM CONSTRUCTION CONTRACTOR at his premises at work site. The original copies shall be filed according to the approved procedure and handed over to the EPCM CONTRACTOR /COMPANY along with the notice of completion of the executed systems.

11.5 SOURCE OF MATERIAL

Before starting earthworks, concrete works, paving and road works, the EPCM CONSTRUCTION CONTRACTOR shall provide a complete list of the names and address of all proposed sources of materials like quarries, concrete mixing, or asphalt plants as well as the relevant suppliers: concrete mix design procedure shall be subject to EPCM CONTRACTOR and COMPANY approval.

The EPCM CONTRACTOR shall have the right to conduct preliminary visits or audits to the proposed sources of materials and to lead or assist to periodical audits, as he may deem appropriate. No material shall be delivered to work site without the prior approval of the EPCM CONTRACTOR.

Should any new source of materials or any new supplier be required during the execution of the works, the EPCM CONSTRUCTION CONTRACTOR shall submit to the EPCM CONTRACTOR the request of approval duly in advance along with pre-qualification details and a description of the proposed scope of work.

11.6 SAMPLING & TESTING

For any material for which, according to the subcontract documents, sampling and testing is required, either at the source or at the works site, and either by means of an independent laboratory or by EPCM CONSTRUCTION CONTRACTOR, the following shall apply, according to Inspection and Test Plans.

The sampling shall be done in the presence of all involved parties (i.e., EPCM CONSTRUCTION CONTRACTOR, EPCM CONTRACTOR, and COMPANY). The samples shall be kept in sealed bags and delivered immediately to the laboratory for testing.

The EPCM CONSTRUCTION CONTRACTOR shall arrange a Site Laboratory suitable to execute the required routine tests at work site.

The tests that cannot be carried out in the Site Laboratory shall be done in Independent Laboratories approved by EPCM CONTRACTOR: any supplier required during the execution of the work shall be from COMPANY approved laboratories List. EPCM CONSTRUCTION CONTRACTOR shall submit the list of the proposed Independent Laboratories to EPCM CONTRACTOR for approval.

The EPCM CONTRACTOR shall have the right to lead preliminary visits and/or audits to the proposed Independent Laboratories and to lead or assist to periodical audits. Only ISO/IEC 17025:2017 ‘’General requirements for the competence of testing and calibration laboratories’’ accredited laboratories shall be utilized for testing of any procured Item/material/Equipment and the calibration of testing equipment.

Fees, and cost for testing, inclusive but not limited to, supply of material, transportation, sampling, testing, storing, supplying of specific equipment, written reports, inspection, workmanship, repetition of tests, whether carried out in the country or outside the country, shall be at EPCM CONSTRUCTION CONTRACTOR charge.

EPCM CONSTRUCTION CONTRACTOR shall establish the Onsite Testing Laboratory for Instrumentation and for Civil as stated in the respective Project Specification. EPCM CONSTRUCTION CONTRACTOR shall submit the dossier containing the list of items, calibration details, technician details and their CVs for Approval.

11.7 RECEIVING INSPECTIONS

EPCM CONSTRUCTION CONTRACTOR shall inspect all incoming materials, equipment for Overage, Shortage & Damage (OSD) and for conformance to all applicable requirements, including the vendor inspection release with verified Material Test Reports (MTR). Data concerning receiving inspections shall be included on the Monthly Materials Status Report. Soft copies of material certificates shall be hyperlinked to heat number and must be kept available in the database for review.

EPCM CONSTRUCTION CONTRACTOR shall submit OSD report to EPCM CONTRACTOR & COMPANY on a weekly basis. For ex-stock material quality control management, refer and comply to AGES-GL-13-001: CONTRACTORs QA/QC Requirements. The company reserves the right to add additional tests including NDE, destructive tests and corrosion tests on material procured from stockiest.

Inspection shall be done by EPCM CONSTRUCTION CONTRACTOR, approved EPCM CONSTRUCTION CONTRACTOR inspectors and records shall be kept for evidence. Material test certificates and other related records shall be available. All records shall be uploaded in the electronic database for quick retrieval at any point of time to verify the inspection status of material. EPCM CONSTRUCTION CONTRACTOR to use special software for materials management. EPCM CONSTRUCTION CONTRACTOR shall submit detailed procedure for Material Management and Preservation to EPCM CONTRACTOR/ COMPANY.

12.0 EXECUTION OF INSPECTION AND TESTING

The EPCM CONTRACTOR & COMPANY shall have the right, always, to examine, inspect, measure, and test the works and the materials.

For this purpose, the EPCM CONTRACTOR, the COMPANY and/or any competent authority shall have, at any time, full access to all parts of the work site, offices, workshops, and places belonging to EPCM CONSTRUCTION CONTRACTOR and LOWER TIER SUBCONTRACTORS where any part of the WORKS is performed, or materials are being obtained or manufactured.

The inspections and tests listed in the approved Site Inspection & Test Plans shall be carried out and duly documented by the EPCM CONSTRUCTION CONTRACTOR in accordance with the Subcontract Documents, the Technical Documentation, and project specification requirements.

The EPCM CONSTRUCTION CONTRACTOR shall schedule the inspections and tests to ensure the regular execution of the WORKS and the smooth start of pre-commissioning and commissioning activities.

The EPCM CONSTRUCTION CONTRACTOR shall provide all necessary supervision and inspection during the execution of PROJECT activities with sufficient competent quality team as per the requirements of AGES-GL-13-001 AND APPENDIX-02.

EPCM CONSTRUCTION CONTRACTOR shall not waive the requirement of any Quality Plan or ITP previously accepted by EPCM CONTRACTOR & COMPANY representative without the express written consent of EPCM CONTRACTOR, COMPANY Representative. EPCM CONSTRUCTION CONTRACTOR shall maintain a Log for all waivers from ADNOC GAS or EPCM CONTRACTORQA/QC and inspection requirements obtained during all phases work. This log should indicate the status of waiver, a brief description and details of the applicable purchase order and equipment or construction process. EPCM CONTRACTOR'S Project Quality Manager shall be delegated the unequivocal authority and direct access to EPCM CONTRACTOR senior management to instigate a cessation of any works that are non-conforming.

12.1 MEASURING EQUIPMENT

All EPCM CONSTRUCTION CONTRACTOR & their subcontractors measuring and testing equipment to be used for the inspections and tests shall be checked, calibrated, and maintained continuously calibrated, according to the approved procedures and ISO 9001:2015 or ISO/TS 29001:2010 requirements.

Inspection equipment shall be attested, calibrated, and certified by independent, recognized and accredited testing laboratory (as per ISO / IEC 17025 Competence of Testing and Calibration Laboratories and shall be in good condition and properly maintained.

EPCM CONSTRUCTION CONTRACTOR shall submit its procedure for calibration and measuring equipment, that will be subject to CONTRACTOR and/or COMPANY approval. This procedure shall comply with ISO 9001:2015 requirements and shall describe how the measuring and testing equipment shall be calibrated, how often they will be recalibrated and the filing of the calibration reports/calibration list.

The EPCM CONSTRUCTION CONTRACTOR shall identify, maintain, control, adjust, and calibrate tools, gauges, instruments and other measuring and testing devices used for activities controlling quality. Calibration shall be performed by recognized third party agencies at established periods as specified in the most stringent of COMPANY standards, other applicable standards, or manufacturer recommendations. If no standards apply, the calibration cycle shall not exceed six months or less according to the type of equipment and sensitivity of the tests.

EPCM CONSTRUCTION CONTRACTOR shall provide backup equipment when the primary equipment is being calibrated or tested. EPCM CONSTRUCTION CONTRACTOR shall verify calibration of measuring and testing devices periodically in accordance with PROJECT specific procedure. EPCM CONSTRUCTION CONTRACTOR personnel performing verification testing/calibrations shall be trained and certified as per applicable international or

manufacturer requirements.

EPCM CONTRACTOR shall periodically verify calibration of measuring and testing devices of EPCM construction CONTRACTOR.

The calibration list and the calibration records shall be stored by the EPCM CONSTRUCTION CONTRACTOR in the EPCM CONSTRUCTION CONTRACTOR's QC at the work site and shall be made available to EPCM CONTRACTOR and COMPANY upon request., refer to APPENDIX 3 for calibration requirements.

12.2 REQUEST FOR INSPECTION

When requested by the approved Site Inspection and Test Plan and before any work is covered up or put out of sight, the EPCM CONSTRUCTION CONTRACTOR shall submit to the EPCM CONTRACTOR the relevant Request for Inspection at least 48 hours in advance refer to project site inspection and coordination procedure PP-30201_50150_LT_HE-GG-000-51-00-008.

When the inspections and tests must be witnessed by external party as Vendor local representative or Licensor local representative, the Inspection Notice shall be submitted to the EPCM CONTRACTOR by EPCM CONSTRUCTION CONTRACTOR at least seven working days in advance, unless any more stringent request by COMPANY.

The Request for inspection shall contain all the necessary information such as, but not limited to, the type of test, the location, the date and the hour, the sequential number of the test, the reference to the applicable codes and standards, the reference drawings.

Request for Inspection template shall be agreed and unified for all the project & shall comply to EPCM and COMPANY procedures.

The participation of the EPCM CONTRACTOR or the COMPANY or any independent inspection agency, the inspection and test shall not release the EPCM CONSTRUCTION CONTRACTOR from any of his obligations and responsibilities under the CONTRACT.

The latest revisions of Drawings, PID, isometrics, installation drawing shall be available at site during the inspection.

12.3 INSPECTION AND TEST RECORDS/CERTIFICATES

EPCM CONSTRUCTION CONTRACTOR shall submit its procedure for management of quality records at site, that will be subject to EPCM CONTRACTOR and/or COMPANY approval.

The EPCM CONSTRUCTION CONTRACTOR shall prepare and fill in all the documents necessary to certify the execution of the inspections and tests and the relevant results.

To prepare the aforesaid documentation the EPCM CONSTRUCTION CONTRACTOR shall use, & will be requested to prepare, the QC forms required and shall be aligned with EPCM CONTRACTOR /COMPANY.

The acceptability or rejection of the inspection/test shall be recorded on the Request for Inspection model submitted to the EPCM CONTRACTOR. The reason being the fact that rejected inspections on total requested is a Key Performance Indicator (KPI).

The original copy of each document shall be stored by the EPCM CONSTRUCTION CONTRACTOR in the EPCM CC's QC office at the work site. All the aforesaid documents shall be filed according to the approved procedure.

The original copies shall be handed over by the EPCM CONSTRUCTION CONTRACTOR to the EPCM CONTRACTOR along with the notice of completion of the relevant portion of works (i.e., systems or subsystem).

12.4 EVALUATION & ACCEPTANCE OF RESULTS & NONCONFOMITY

The EPCM CONSTRUCTION CONTRACTOR shall evaluate the results of the inspections and tests to assess whether the same complies with the project requirements & applicable codes.

Whenever the results of the inspections and tests do not comply with the applicable codes and specifications (i.e., there is a "nonconformity") the EPCM CONSTRUCTION CONTRACTOR shall proceed as follows.

First, the EPCM CONSTRUCTION CONTRACTOR shall verify the correctness of the procedure used for testing, the instrument calibration and setting, the qualification of the operator, etc., to be sure about the nonconformity discovered.

When the nonconformity is confirmed, the EPCM CONSTRUCTION CONTRACTOR shall issue a nonconformity report proposing the rejection or the correction of the item concerned. EPCM CONSTRUCTION CONTRACTOR shall submit its procedure for non-conformity report, that will be subject to EPCM COMPANY and COMPANY approval.

The EPCM CONTRACTOR shall review the EPCM CONSTRUCTION CONTRACTOR evaluation of the results of the inspections and tests and to accept or reject EPCM CONSTRUCTION CONTRACTOR proposal for the correction of the nonconformity/rejection.

The participation of the EPCM CONTRACTOR or the COMPANY or any independent inspection agency in the evaluation process and/or their acceptance of the results of the inspection and test shall not release the EPCM CONSTRUCTION CONTRACTOR from any of its obligations and responsibilities under the SUBCONTRACT.

All repairs and reinstatements carried out in accordance with the approved procedures shall not be treated as nonconformity.

Might the EPCM CONSTRUCTION CONTRACTOR delay or refuse to issue a nonconformity report the EPCM CONTRACTOR reserves to proceed anyway and to charge the cost related to any impact on prosecution of activities.

Any nonconformance detected during inspection activities shall be processed and corrected

using the Project Nonconformities management procedure, approved by COMPANY NO. PP-30201_50150_LT_HE-GG-000-51-00-003

EPCM CONSTRUCTION CONTRACTOR shall submit its nonconformance procedure for review and approval to EPCM CONTRACTOR prior to the start of the work.

The procedure shall describe:

•the reporting system of the non-conformities initiated by EPCM CONSTRUCTION CONTRACTOR.

•the sequential numbering system.

•the non-conformities logging system and distribution.

•the responsibilities and the description of the management of non-conformities within the EPCM CONSTRUCTION CONTRACTOR organization (issuing, evaluation/approval, closure).

•the segregation of non-conforming items.

EPCM CONTRACTOR shall issue NCR for all violations by EPCM CONSTRUCTION CONTRACTOR to PROJECT requirements.

All NCRs shall be properly reviewed and concluded by root cause analysis and corrective actions implemented within the quality system.

EPCM CONSTRUCTION CONTRACTOR shall maintain a summary of open NCRs and submit it, on weekly basis, to EPCM CONTRACTOR.

NCR register & closed out NCR's; CAR register & closed out CAR's; Change Management register and closed out & signed request forms. (Technical Queries - TQ, Deviation Request – DR and Concession Request - CR, etc.) and related logbook, to be part of the final dossier BY EPCM CONSTRUCTION CONTRACTOR.

As part of its own Quality System, COMPANY will operate its own non-conformance reporting to EPCM CONSTRUCTION CONTRACTOR. When a COMPANY non-conformance report is issued, EPCM CONSTRUCTION CONTRACTOR shall immediately acknowledge receipt by signature on the original non-conformance report form. On receipt of COMPANY non-conformance reports, EPCM CONSTRUCTION CONTRACTOR shall prepare a corrective action plan within 72 hours.

Where such non-conformances are identified as major the affected work shall be stopped. Where the COMPANY has concerns over the quality of WORK, it shall have the right to re-test. Where WORK is covered over, the re-test shall include removal of covering.

No WORK shall be covered up without the approved procedures and ITP intervention points being closed and signed, therefore MECHANICAL COMPLETION shall not be confirmed until all QC activities are signed off.

12.5 STATISTICAL ANALYSIS

Both the positive and the negative results of the inspections and tests performed shall be recorded in the Request for Inspection Logbook that shall be regularly submitted to EPCM CONTRACTOR, weekly as a minimum.

The aforesaid Logbook shall indicate the discipline, the total number of Request for Inspection issued to date and how many have been accepted, cancelled, or rejected with the respective percentage over the total.

The failure percentages for each discipline shall be monitored during the execution of the works to identify the prevailing trend and adopt, as early as possible, adequate corrective measures.

13.0 AUDIT PROGRAM

During the entire course of the WORKS the EPCM CONTRACTOR have the right to audit any aspect of the EPCM CONSTRUCTION CONTRACTOR's Quality Assurance System.

EPCM CONTRACTOR will perform audits to EPCM CONSTRUCTION CONTRACTOR at site according with the document PP-30201_50150_LT_HE-GG-000-51-00-002 PROJECT Audit procedure.

By way of example, and without limiting in any way the generality of the foregoing, the EPCM CONTRACTOR will carry out audits to verify:
-the correct identification and the proper storage conditions of the materials.
-the qualification and identification of the personnel performing special processes as defined by ISO 9001: 2015 and/or ISO/TS 29001:2010.
-the usage of the latest approved construction drawings and documents.
-the implementation of the approved Site inspection and test Plans.
-the execution of the required measurements, inspections, and tests.
-the identification, correct calibration and setting of measuring equipment.
-the compliance with the approved Method Statements.
EPCM CONSTRUCTION CONTRACTOR shall submit its procedure for audit at site, that will be subject to EPCM CONTRACTOR and COMPANY approval.

EPCM CONSTRUCTION CONTRACTOR shall submit its detailed audit schedule and checklists to COMPANY including audits of SUBCONTRACTOR's / Vendors. Any rescheduling of the audits can be done only with formal approval of the COMPANY representative. Follow up audits shall be conducted at regular intervals on key areas in accordance with the EPCM CONTRACTOR audit program duly approved by the COMPANY. The internal auditor shall have enough experience in auditing and certified by an international auditor's certification authority (e.g., IRCA or Equivalent).

14.0 PRESERVATION

EPCM CONSTRUCTION CONTRACTOR shall issue a material handling and preservation procedure for EPCM CONTRACTOR and COMPANY approval with one month of the agreement.

EPCM CONTRACTOR, together with EPCM CONSTRUCTION CONTRACTOR, shall maintain and implement procedures for preservation and preventative maintenance of materials and equipment from the time of arrival at work site until they are turned over to COMPANY.

These procedures shall include requirements for the use and application of corrosion preventives.

Any special handling, preservation, storage, cleaning, packaging, and shipping requirements will be identified, particularly those that must be accomplished by qualified individuals in accordance with predetermined inspection instructions.

EPCM CONTRACTOR shall ensure that, upon receipt at the Work Site, EPCM CONSTRUCTION CONTRACTOR shall examine all equipment shipped with protective covers or plugs over openings and applied corrosion preventive coatings on exterior machined and traded surfaces.

The program for preventive maintenance of equipment shall be in accordance with the manufacturer's recommendations and shall provide a sufficient level of maintenance acceptable for warranty purposes. EPCM CONSTRUCTION CONTRACTOR shall maintain records to verify the maintenance of equipment performed during construction and commissioning in accordance with procedures and method statements agreed with EPCM CONTRACTOR /COMPANY. These records shall be then turned over to EPCM CONTRACTOR as part of EPCM CONSTRUCTION CONTRACTOR documentation and records prepared for mechanical completion.

15.0 REPORTING

EPCM Construction CONTRACTOR shall prepare and submit to EPCM CONTRACTOR on a Weekly & monthly basis evidence to demonstrate effective implementation of the QMS.

As a minimum, EPCM CONSTRUCTION CONTRACTOR shall include the following items:

-Summary of the plans, procedures, work instructions, method statements developed and respective review, approval, and issue status

-Summary of the audits carried out reported against the audit schedule

-Listing of findings and corrective actions per each audit, the status and the planned dates for follow up and close out

-Nonconformance reports issued during construction: number issued, closed out, pending and planned dates for follow up and close out

-Status of NCR and corrective/preventive actions

-Summary of inspection and tests carried out on site in reference to the activities listed in ITP

-Listing and disposition of all defective or rejected material or equipment received at jobsite

-Welding rejection rate reported on per joint basis, for piping, pipeline, structural and alloy welds.
-Status of preservation and preventive maintenance program
-KPI which agreed at start of activities for monitoring project progress and control of project performance, the KPI s shall be in line with the COMPANY specification requirements and
-Narrative section for process improvement
Several of these above-mentioned items shall be submitted to EPCM on weekly basis, as stated in the table above for the EPCM CONSTRUCTION CONTRACTOR submissions.
EPCM CONSTRUCTION CONTRACTOR to establish, maintain and manage integrated database for materials management and certification, inspection report preparation and certification progress for all disciplines, welding, NDE, piping test packs, punch list, sub system and system mechanical completion. The database shall be developed based on oracle or equivalent software and shall generate various kinds of output reports as required by the EPCM CONTRACTOR & COMPANY quality management team

16.0 FINAL DOSSIERS BY EPCM CONSTRUCTION CONTRACTOR

EPCM CONSTRUCTION CONTRACTOR to produce a Final Dossier Programme for EPCM & COMPANY review and approval. A preliminary version to be submitted by EPCM CONSTRUCTION CONTRACTOR within 30 days by KOM. Reached out the 50% of the work progress, to be submitted for EPCM Review and approval the Final Dossier issuing Programme.

The Final dossier shall be uploaded in ECMS Live link for easy access and retrieval. Digitized RT films in JPG formats shall be hyperlinked along with Radiography reports for each line no and joint no for submission to EPCM CONTRACTOR & COMPANY BY EPCM CONSTRUCTION CONTRACTOR.
Test pack documents & Loop folders shall be scanned and hyperlinked for each unit and sub-unit for submission to EPCM CONTRACTOR & COMPANY. Material test certificates for each heat no shall be hyperlinked and submitted to EPCM CONTRACTOR & COMPANY. During the life of the project, all inspection reports, critical documents, and laboratory reports shall be scanned (PDF searchable) and kept in server.

17.0 QUALITY MEETINGS

EPCM CONTRACTOR shall initiate a quality kick-off/ Pre-production meeting with all concerned PARTIES including EPCM CONSTRUCTION CONTRACTOR and COMPANY to ensure full understanding on the execution SOW, quality requirements and the level of required control on the construction activities.

Weekly QA/QC Coordination meetings shall be conducted between EPCM CONCTRUCTION CONTRACTOR’s and EPCM CONTRACTOR’s Quality teams, along with COMPANY.

This meeting shall focus primarily on look-ahead schedules and quality related matters and continuous opportunities of improvement. EPCM CONTRACTOR should review the EPCM CONSTRUCTION CONTRACTOR's look-ahead schedule to assist in the planning of coverage of the WORK activities. Emphasis should be placed on the start of new activities to prevent construction problems, and on the receipt of major equipment for early detection of potential deficiencies.

18.0 APPENDICIES

Appendix-1: Minimum Requirements for Number of Quality Personnel
Appendix-2: Minimum Requirements for Qualification of Quality Personnel
Appendix-3: Control of construction process & calibration.
Appendix-4: Document issuance responsibility matrix.

APPENDIX-1

MINIMUM REQUIREMENTS FOR NUMBER OF QUALITY PERSONNEL

(NO. OF PAGES: 04)

This table applicable to EPCM CONSTRUCTION CONTRACTOR

Table-1 Minimum Contractor's Quality Organization. (Onshore /Offshore projects).			
Project Value in USD (EPC Value)	**Below 50M USD**	**From 50M to 200M USD**	**Above 200M USD**
Overall Project Quality Manager (PQM)		Full time PQM to manage all Project phases QA/QC	Full time PQM to manage all Project phases QA/QC
Construction Quality Manager (CQM)		Full time CQM to manage all construction Site QA/QC	Full time CQM to manage all construction Site QA/QC
Sr. QC Engineer	Full time Sr. QC Eng. to manage all Project QC functions	Full time Sr. QC Eng. to manage all Project QC functions	Full time Sr. QC Eng. per each discipline as applicable E&I, CIVIL, Material Mechanical; etc.
Sr. QA Engineer		One full time to manage all Project phases QA activities and Audits.	One full time to manage all Project phases QA activities and Audits.
Lead Procurement QC Eng.			One Full time Lead Procurement QC Eng, to manage Procured equipment /Materials QC.
Procurement QC Coordinator	One full time to coordinate all Procurement QC activities	One full time to coordinate all Procurement QC activities	One or more full time to coordinate all Procurement QC activities.
QC Engineer (Procurement)	Min. one Full time QC Eng. to manage all Procurement QC	Min. one full time QC Eng. Per discipline to Manage all Procurement QC.	Min. one full time QC Eng. Per discipline to manage all Procurement QC
QC Engineer (Construction)	One	One per discipline	One or more per discipline based on project complexity
QA Engineer	One QA Engineer to manage all Project QA functions and audits.		Contractor to decide based on the complexity of the Work
Sr. Painting/ Coating Engineer	One part time	One full time	One Full time Sr. Painting Engineer
Sr. WELDING ENGINEER	One part time Welding Engineer	One full time Sr.Welding Engineer	One Full time Sr. Welding Engineer
NDT Level III or Equivalent	One part time	One full time	One full time
QC supervisor			One /discipline for each 6 Inspectors onshore & 1 for offshore.

Table-2 Shop /Construction site QC Inspectors requirements for Onshore PROJECTs.

Discipline	Contractor	EPCM construction contractor
Civil/Structural/GRE		
Earth Works & Paving	1 per 5 Sub-Contractor Inspectors	1 per 50 Workmen
Concrete Foundation	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
Buildings	1 per 5 Sub-Contractor Inspectors	1 per 50 Workmen
Steel structural(site or shop)	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
GRE Works*	1 per 5 Sub-Contractor Inspectors	1 Per 10 Workmen
HVAC & MEP	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
Refractory & Fire proofing	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
Mechanical/ Piping		
Receiving Inspector	1 at each Site Warehouse	1 at each Site Warehouse
Rotating & Mechanical Inspector	1 per 5 Sub-Contractor Inspectors	1 per 15 Workmen
Piping	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
Valve testing	1 per Valve testing Location	1 per Valve testing Location
Coating and Insulation	1 per 8 Sub-Contractor Inspectors	1 per 3 crews per Site
Test Pack Coordinator	As per Requirements. (Minimum One)	Minimum One

Table-2 Shop /Construction site QC Inspectors requirements for Onshore PROJECTs.

Discipline	*Contractor*	*EPCM construction contractor*
Welding		
Receiving Inspector	1 at each Site Warehouse	1 at each Site Warehouse
Field and shop Welding	1 per 5 Sub-Contractor Inspectors	1 per 15 Workmen
NDT & PMI & PWHT / Hardness Coordinators	As per Requirement (Minimum One) and based on NDE volume and locations, more are required.	Minimum One in addition to RT film interpreters. More are required depending upon volume of NDE and spreads/locations
Welding Engineer*	1 dedicated to PROJECT	1 per Work Site
NDT Level III Engineer	1 dedicated to PROJECT	1 per Work Site
PIPELINE WOKS		
Receiving Inspector	1 at each Site	1 at each Site Warehouse
Stringing & Bending	1 per PROJECT	2 per PROJECT.
Field & Repair Welding	1 per each Crew(Onshore)	2 per Crew mainline & Tie ins. I per Repair Crew
NDT	1 Per each Crew – At Site for AUT. Rest – At Site office – As required. For RT, separate film interpreters are required.	1 Per each Crew – At Site for AUT Rest – At Site office – As required. For RT, separate coordinator and film interpreters arerequired

Table-2 Shop /Construction site QC Inspectors requirements for Onshore PROJECTs.

Discipline	***Contractor***	***EPCM construction contractor***
Backfilling (onshore only)	1 for each Crew	1 for each Crew
Coating Inspector	1 for each Crew	1 for each Crew
Hydro test Coordination & Document Review	1 Quality Engineer	1 per Test Crew
Cathodic Protection	1 Per PROJECT	1 Per PROJECT
ELECTRICAL / INSTRUMENTATION		
Electrical and instruments material Receiving Inspector	1 at each Site Warehouse	1 at each Site Warehouse
Electrical / Cathodic Protection	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
Instrumentation	1 per 5 Sub-Contractor Inspectors	1 per 25 Workmen
Calibration of Valves	1 per Crew	1 per Crew
Telecom	1 per Site	1 per Site

* All site GRE installation and testing shall be witnessed, supervised by manufacturer technical representatives on full time basis. All inspection records to be endorsed for good workmanship.

- For every 5 CONTRACTOR Inspectors or more per discipline, there shall be one QC engineer and QC supervisor in that discipline. Similarly for every 5 SUBCONTRACTOR Inspectors or more per discipline, there shall be one QC engineer and one QC supervisor in that discipline
- A minimum of one (1) QC engineer and one (1) QC Inspector will be assigned by CONTRACTOR/SUBCONTRACTORS for each position above if the number of workers / labors / welders / technicians and QC Inspectors performing the activities is below the specified ratio.

APPENDIX-2

MINIMUM REQUIREMENTS FOR QUALIFICATION OF QUALITY PERSONNEL
(NO. OF PAGES: 07)

QUALIFICATION REQUIREMENTS

PROJECT Quality Manager

- Bachelor of mechanical or metallurgical engineering, or other relevant engineering degree.
- Certified ISO 9001 Lead Auditor.
- Having a total of min. 20 years with specific experience as a QA / QC manager for min. 10 years in the Oil & Gas PROJECTs.
- Good command of PROJECT Quality auditing.
- Strong personality and leadership skills.
- Fluent English lingual skills.

Construction Quality Manager:

- Bachelor of mechanical or metallurgical engineering, or other relevant engineering degree.
- Certified ISO 9001 Lead Auditor.
- Having a total experience of min. 20 years with specific experience as a QA / QC manager for min. 10 years in the Oil & Gas PROJECTs.
- Relevant background in welding, NDT, and coating / painting.
- Strong personality and leadership skills.
- Fluent English lingual skills.

PROJECT Quality Assurance Manager:

- Bachelor of mechanical or metallurgical engineering, or other relevant engineering degree.
- Certified ISO 9001 Lead Auditor.
- Having a total experience of min. 20 years with specific experience as a QA manager for min. 5 years in the Oil & Gas PROJECTs.
- Good command in PROJECT Quality auditing.
- Strong personality and leadership skills.
- Fluent English lingual skills.

QC Senior Engineer

- Bachelor of mechanical or metallurgical engineering, or other relevant engineering degree.

- ISO 9001 Lead Auditor.
- ASNT Level II in RT, UT, MT, and PT.
- CSWIP 3.2 or AWS-CWI or another acceptable equivalent (approved by COMPANY).
- Having a total experience of min. 15 years in the Oil & Gas PROJECTs.
- Strong experience in welding, NDT, and coating / painting.
- Strong supervisory, coordination, and reporting skills.
- Fluent English lingual skills.

QC Engineer

- Bachelor of mechanical or metallurgical engineering, or other relevant engineering degree.
- ISO 9001 Internal Auditor.
- ASNT Level II in RT, UT, MT, and PT.
- CSWIP 3.1 or AWS-CWI or another acceptable equivalent (approved by COMPANY).
- Having a total of min. 8 years as QC Engineer in the Oil & Gas PROJECTs.
- Hands on experience in welding, NDT, and coating / painting.
- Good command of PROJECT Quality auditing.
- Strong coordination and reporting skills.
- Fluent English lingual skills.

Sr. QA Engineer / QA Engineer

- Bachelor of engineering in a relevant discipline.
- Certified and registered IRCA Lead Auditor in ISO 9001
- Having a total of min. 12 years as QA Engineer with strong background in the Oil & Gas PROJECTs environment.
- Strong knowledge and experience in leading PROJECT Quality audits.
- Strong coordination and reporting skills.
- Fluent English lingual skills.

Welding Engineer

- Bachelor of mechanical or metallurgical engineering with Masters' Degree in welding.
- Alternatively, Bachelor of mechanical or metallurgical engineering with Professional certification in welding engineering (IWE, CWE, or another acceptable equivalent (approved by COMPANY)).

- ASNT Level II in RT, UT, MT, and PT.
- Having a total experience of min. 12 years in welding related activities within the Oil & Gas PROJECT environment.
- Fluent English lingual skills.

Welding Inspector

- Having a total experience of min. 5 years in the Oil & Gas industry (construction and source inspection).
- Engineering Diploma in a relevant discipline.
- CSWIP 3.1 or AWS-CWI or another acceptable equivalent (approved by COMPANY).
- Good English lingual skills.

Sr. Coating / Painting Inspector

- Having a total of min. 10 years in the Oil & Gas industry (construction and source inspection).
- Engineering Diploma in a relevant discipline.
- NACE Level II or CSWIP / BGAS Grade 2, or another acceptable equivalent (approved by COMPANY).
- Good English lingual skills.

Electrical Inspector

- Shall be at least a diploma in Electrical discipline and a minimum of 6 years of Electrical Construction QC experience involving Oil & Gas / Petrochemical PROJECTs.
- Inspectors shall preferably be certified to COMPEX or equivalent requirements.

Instrumentation Inspector

- Shall be at least a diploma in Instrumentation discipline and minimum 6 years of Instrumentation Construction QC experience involving Oil & Gas / Petrochemical PROJECTs.

Coating / Painting Inspector

- Having a total experience of min. 5 years in the Oil & Gas industry (construction and source inspection).
- Engineering Diploma in a relevant discipline.
- NACE Level I / II or CSWIP / BGAS Grade 1 / 2, or another acceptable

equivalent (approved by COMPANY).

- Good English lingual skills.

Civil Inspector

- Shall be at least a diploma in civil discipline and a minimum of 6 years of civil Construction QC experience involving Oil & Gas / Petrochemical PROJECTs.

Plant and Equipment Inspector:

- Inspector shall demonstrate working knowledge of codes such as ASME Sec I, V, VIII and IX. He shall be able to perform all types of equipment inspection such as inspections that require the use of callipers, micrometres, infrared camera, boroscope, pit gauges etc. to ensure continued safe operation; tests and inspections on equipment that has been repaired or modified to assure that the equipment meets the manufacturers and ASME Code requirements; inspections for the installation of hot tap, weld plus ends, repair sleeves, or any welding that is required on equipment while in service; and inspections on vessels and equipment heat treatments. For rotating equipment, the inspector shall have through knowledge in API 686 and familiar with installation, initial alignment, final alignment by dial gages or laser, periodic preservation etc. For static equipment, the inspector shall be familiar with trays installation inspection, laser alignment and levelling, equipment field assembly, box up inspection, preservation etc

Heat Treatment Inspector:

- Inspector shall have five years direct work experience as a heat treatment technician (preferably with some supervisory experience) working directly for an industry recognized heat treatment CONTRACTOR(s). Inspector shall have direct knowledge of all aspects of heat treatment with the types of equipment that will be used on the construction project such as electrical resistance heating elements, induction coils, or gas fired heaters. Related experience in

any of these areas qualifies Inspector to also inspect all types of shop or site erected heat-treating furnaces for batch heat treatment of parts and vessels that are internally fired to stress relieve them. Inspector shall also be thoroughly familiar with the heat-treating requirements of ASME B31.3 and ASME Section VIII Division 1 and AWS D10.10.

HVAC Inspector:

- Inspector shall demonstrate a thorough working knowledge of codes such as the Sheet Metal and Air Conditioning CONTRACTORs National Association (SMACNA) codes and/or equivalent standards

Communication Inspector:

- Inspector shall demonstrate a thorough working knowledge of telecommunications, and be familiar with North American or International Telecommunications Standards; Building Industry Consulting Services International - Telecommunications Distribution Methods Manual (BICSI-TDMM); National Electrical Codes; and National Electrical Safety Codes, or their equivalents

Plumbing Inspector:

- Inspector shall demonstrate a thorough knowledge of codes such as the Uniform Plumbing or equivalent standards.

VENDOR Inspectors

- Having total relevant experience of min. 5 years in the Oil & Gas industry (VENDOR / source inspection).
- Engineering Diploma in a relevant discipline.
- Familiar with Shell DEP's and international codes & standards.
- Professional qualification / certification in the relevant discipline (welding, coating, painting, NDT, …etc.).
- QA/QC Inspectors (GRE /GRP, Piping, Pressure Vessels, Tanks, Civil, Electrical / Instrumentation, Cathodic protection … etc.) Having a total experience of min. 5 years in the Oil & Gas industry (construction and source inspection).

- Engineering Diploma in a relevant discipline.
- Certified to the related discipline in Quality.
- Good English lingual skills.

NDT Technicians

- Radiographer shall be certified to ASNT or CSWIP / PCN level II with 3 years' experience after certification, relevant to Oil & Gas PROJECTs.
- RT film interpreter shall be certified to ASNT ACCP level III, or CSWIP / PCN level II in RT with 5 years' experience after certification, relevant to Oil & Gas PROJECTs.
- Ultrasonic testing technician shall be certified to CSWIP / PCN level II certification with 5 years' experience after certification, relevant to Oil & Gas PROJECTs.
- Other technicians shall have ASNT or CSWIP / PCN level II certification with min. 5years of relevant experience in the Oil & Gas industry
- Alternative equivalent qualifications may also be acceptable, subject to COMPANY approval.

Advanced NDT Inspector - AUT/PAUT/TOFD

- Hold a Degree/Diploma in Engineering or an approved equivalent.
- Shall be qualified in accordance with TWI CWSIP ISO 9712 Level II for AUT Inspector.
- Shall be qualified in accordance with BINDT/PCN ISO 9712 Level II for PAUT / TOFD Inspector.
- Shall have prior experience in the relevant method and shall have worked in 3 prior PROJECTs in a similar capacity.

RT film digitization technicians:

- The technicians shall possess valid certificate in the process of digitization and their training details and records shall be submitted to COMPANY for review. The technicians shall demonstrate their knowledge and ability in the film digitization, storing and navigation processes to obtain the same degree of quality of the original film. Only ASNT Level II RT technicians

with film interpretation certification shall be engaged for this activity.

Radiography source storage and handling:

NDE agency shall submit the safe storage and handling procedure, location approval to COMPANY for review.

As a minimum, the Storage facility should be a lockable room or a purpose-built unit that provides protection and Safety. It should be designated as a controlled area or supervised area, where appropriate, and should be:

a) Resistant to fire

b) Away from any fire, explosion, or corrosion hazards

c) Kept locked with the keys held only by authorized personnel. A warning notice including the radiation symbol (trefoil) should be displayed on the door

APPENDIX-3
CONTROL OF CONSTRUCTION PROCESS
(NO. OF PAGES: 03)

Special Process	Work Procedure Approval (A) or Review (R) required	Qualification Procedure Approval Required	Personnel Qualification Approval Required
Welding	A	Yes	Yes
Non-destructive Testing / Examination	A	Yes	Yes
Cable splicing / termination	R	Yes	Yes
Alloy verification	A	Yes	Yes
Coating	A	Yes	Yes
Fireproofing	A	Yes	Yes
Concrete Mix design	A	Yes	Yes
Cable and conduit seal installation	R	No	No
Gasket specification and installation	R	No	No
Concrete installation	R	No	No
Structural Bolt tensioning	R	No	No
Gap Control for Socket weld and seal weld threaded fittings	A	Yes	Yes
Hydro-testing	A	No	No
Heat treatment	A	No	No
Hot Tap Procedures (Calculation sheet and Check Lists)	A	Yes	Yes

Refractory installation	A	No	No
Hardness test	A	NO	NO
Destructive testing and civil material testing	R	NO	NO
Loop testing	A	No	Yes
Flange Management	A	YES	YES
Baseline wall thickness survey	A	YES	YES
CP System	R	Yes	Yes
GRE Piping (BONDING, LAMINATION & Procedures)	A	Yes	Yes
Pipeline laying	R	Yes	Yes
Pipeline coating	R	Yes	Yes
Electrical testing	R	Yes	Yes
Instrumentation	R	Yes	Yes

Devices CALIBRATION CHART – REQUIREMENT

Description	Calibration Frequencies	Tolerance
Primary Standard		
Gage Blocks	5 Years	Factory Stds.
Test Bars/Rods	5 Years	Factory Stds.
Proving Ring	5 Years	Factory Stds.
Mercury Thermometer	1 Year	+ 2° F.
Laboratory Hardness Testing Equipment	1 Year	Factory Stds
Load		
Load-Set Machines	3 Months	2%
Dynamometers	3 Months	2%
Tensile Test Machines	1 Year	1%
Snubber Test Ring	6 Months	2%
Impact Testing Equipment	1 Year	1%
Temperature		
Thermocouple	3 Months	Greater of 2% or 25°F
Temperature Recording Instrument	6 Months	1% of Chart
Optical Pyrometer	6 Months	1% of Meter
Contact Pyrometer	6 Months	1% of Meter
Electrical		
Tong Tester	6 Months	1% of Meter
Volt-Ohm-Amp Meter	6 Months	2% of Meter
Potentiometer	18 Months	2% of Meter
Welding Machines validation	Once in a year	As per BS 7570
Pressure		
Welding Gas Proportions	3 Years	2%
Pressure Gauge/Recorders	15 days/ 3Months	Factory Stds.
Dead Weight Tester	1 Year	1%
NDE		
Magnetic Particle Equipment - Yoke	6 Months	(ASME Sec. V)
Magnetic Particle Equipment - Prod/Coil	6 Months	2% of Meter
Ultrasonic Equipment*	Each Use	
Portable Hardness Testing Equipment*	(Note 1)	
Portable Chemistry Analyzers*	(Note 1)	
Densitometer*	Each Use	
Dimensional		
Outside Micrometers	3 Months	.002"
Inside Micrometers	3 Months	.001"
Ultrasonic Thickness Gauge*	Each Use	8%
Paint Thickness Gauge - Magnetic	3 Months	8%
Paint Thickness Gauge – Electronic	3 Months	Factory Standards
Electronic Level	1 Year	

Note - 1 these devices shall be verified against two (2) known standards prior to use.
* This equipment does not require a calibration sticker.
Calibration laboratory shall be accredited to ISO 17025 by UKAS or IAS or DAC or ENAS and the scope of accreditation shall cover the specific procedures applicable for calibration of each instrument. Calibration of any other instrument not listed in this specification shall be treated in accordance with UKAS standards.

Appendix-4

Document issuance matrix.

Title	EPCM CONSTRUCTION CONTRACTOR ISSUANCE (I)	EPCM CONTRACTOR Issue (I)/Review (R)/ Approval(A)	COMPANY Issue (I)/Review (R)/ Approval(A)
EPCM CONTRACTOR Construction Quality Plan		I	A
EPCM CONSTRUCTION CONTRACTORS Quality Plan	I	A	A
EPCM CONTRACTOR Nonconformities management procedure		I	A
EPCM CONSTRUCTION CONTRACTORS Nonconformities management procedure	I	A	R
EPCM CONSTRUCTION CONTRACTORS Management of quality records procedure	I	A	A
EPCM CONTRACTOR Project Audit program and procedure		I	A
EPCM CONSTRUCTION CONTRACTORS Audit procedure	I	A	R
EPCM CONSTRUCTION CONTRACTORS Procedure for Calibration of Measuring Equipment	I	A	A
EPCM CONSTRUCTION CONTRACTORS Procedure for Filing documentation	I	A	A
EPCM CONSTRUCTION CONTRACTOR Site General Preservation Procedure	I	A	R
EPCM CONTRACTOR Site Quality Audits to EPCM CONSTRUCTION CONTRACTORS		I	R

EPCM CONSTRUCTION CONTRACTORS NCR (Reports)	I	A	A
EPCM CONSTRUCTION CONTRACTOR Inspection and test plans but not limited to the following : -ITP for Early site preparation, -ITP for Civil works. -ITP for piping prefabrication -ITP for piping installation. -ITP for steel fabrication and installation. -ITP for pipe line laying. -ITP Painting application -ITP for fireproofing. -ITP for pipe line coating -ITP covering all electrical parts and electrical activities. -ITP covering Instrumentation part and instrumentation installation. ITPs covering pre-commissioning SOW discipline wise.	I	A	A
QC Procedures but not limited to the below: • concrete mix design • Inspection Control System for Welding Activities • Fabrication of Piping Spool at Workshop • Assembly of Piping Spool at Field • Procedure for the Updating of the Welds Status • Procedure for Storage and Handling of Welding Consumables • Procedure for Visual Examination			

• Procedure for Liquid Penetrant Examination • Procedure for Magnetic Particle Examination • Procedure for Radiographic Examination • Procedure for Ultrasonic Examination • Procedure for Dimensional Control • Procedure for Post Weld Heat Treatment of Carbon and Low Alloy Steel Piping • Procedure for Hydrotest • Procedure for HDPE piping • Procedure for GRE piping • Procedure for Cathodic Protection • Erection of Steel Structure • Electrical and Instrumentation Works • Surface Preparation and Painting Control Procedure	I	A	A
EPCM CONSTRUCTION CONTRACTOR QA/QC Personnel Qualification	I	A	A
Welding Procedure Qualification Records (PQR)	I	A	A
Welding Procedure Specifications (WPS),	I	A	A
Welder Performance Qualification Record and relevant Welder List	I	A	A
Non-Destructive Test (NDT) Personnel List with the relevant certification and Curriculum Vitae	I	A	A
NDT/PWHT procedures	I	A	A
All construction Method statements	I	A	A